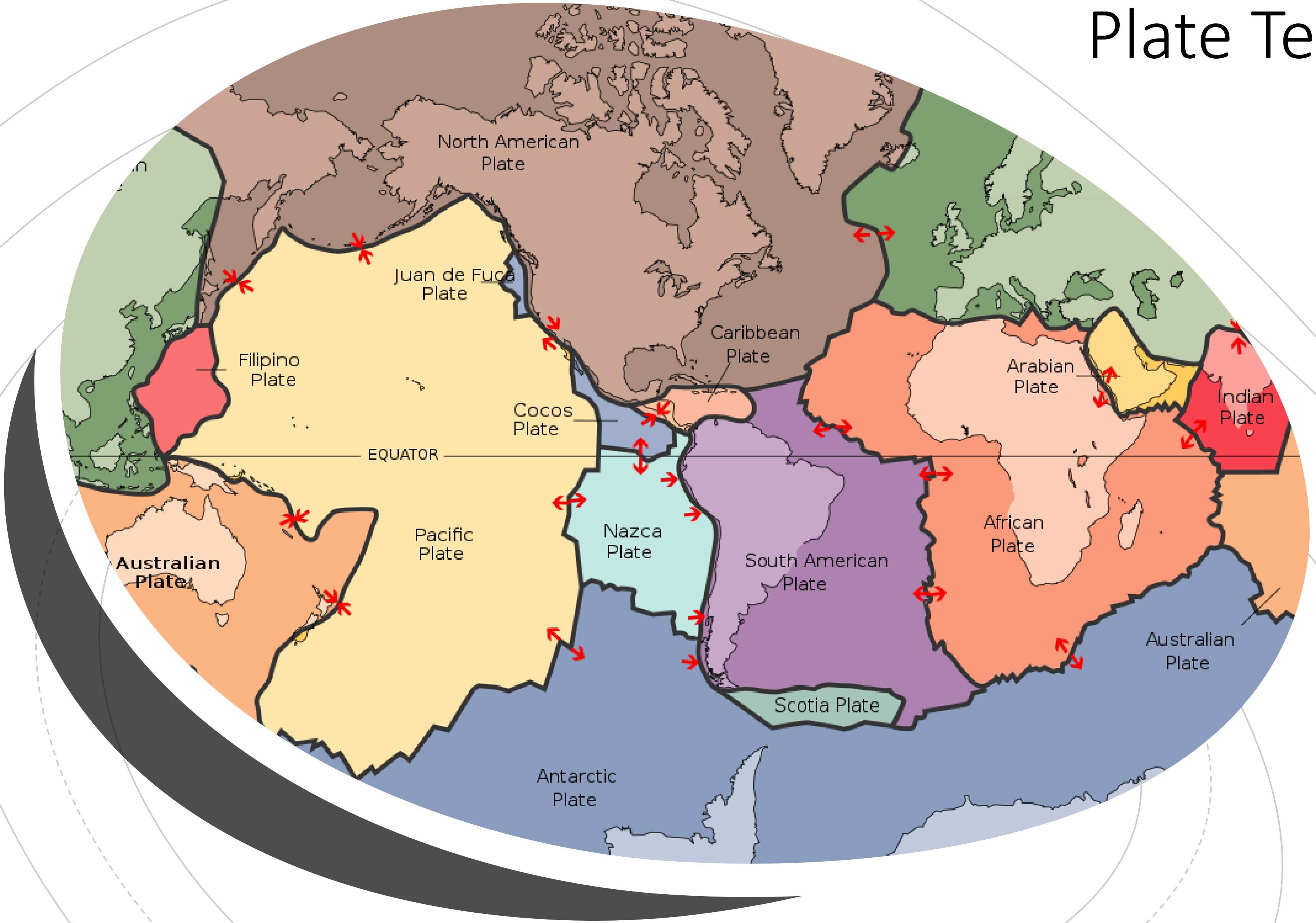


Plate Tectonic Paradigm

Reminders:

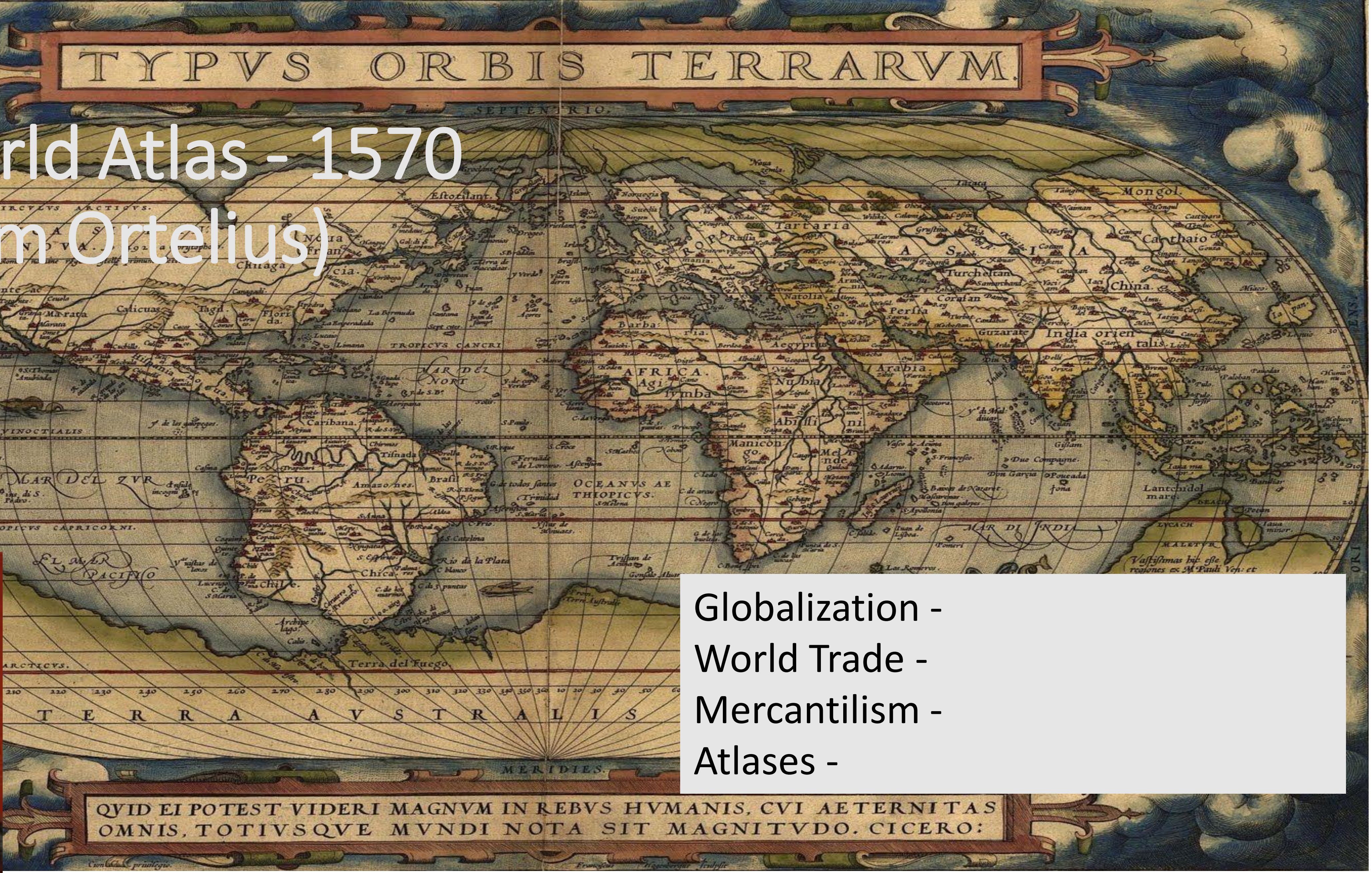
Module 1 available today! – due Jan 30

Course outline quiz bonuses only available if completed before end of month!

Next week: Structure of the Earth

Today: Key Steps to Plate Tectonic Theory

- North America and Europe can be ‘fitted together’: 1600 (Francis Bacon)
- An ‘ancestral continent’: 1858 (Snider Pellegrini)
- ‘Earth’s Plan’: 1910 (Frank Taylor)
- **‘Continental Drift**’ and Pangea: 1915 (Alfred Wegener)
- Mantle convection currents: 1928 (Arthur Holmes)
- Mid-ocean ridges and trenches: 1957 (Marie Tharp and Bruce Heezen)
- Sea floor spreading’: 1962 (Harold Hess)
- 1965 ‘Wander paths’ for continents (Ted Irving) and ‘magnetic stripes’ on the ocean floors (Fred Vine)
- Computer reconstruction of Pangea: 1965 (Edward Bullard)
- Transform faults and hot spots: 1965 (Tuzo Wilson)
- **Plate tectonics** 1968 (Tuzo Wilson)
- Supercontinent cycle 1974 (John Dewey)



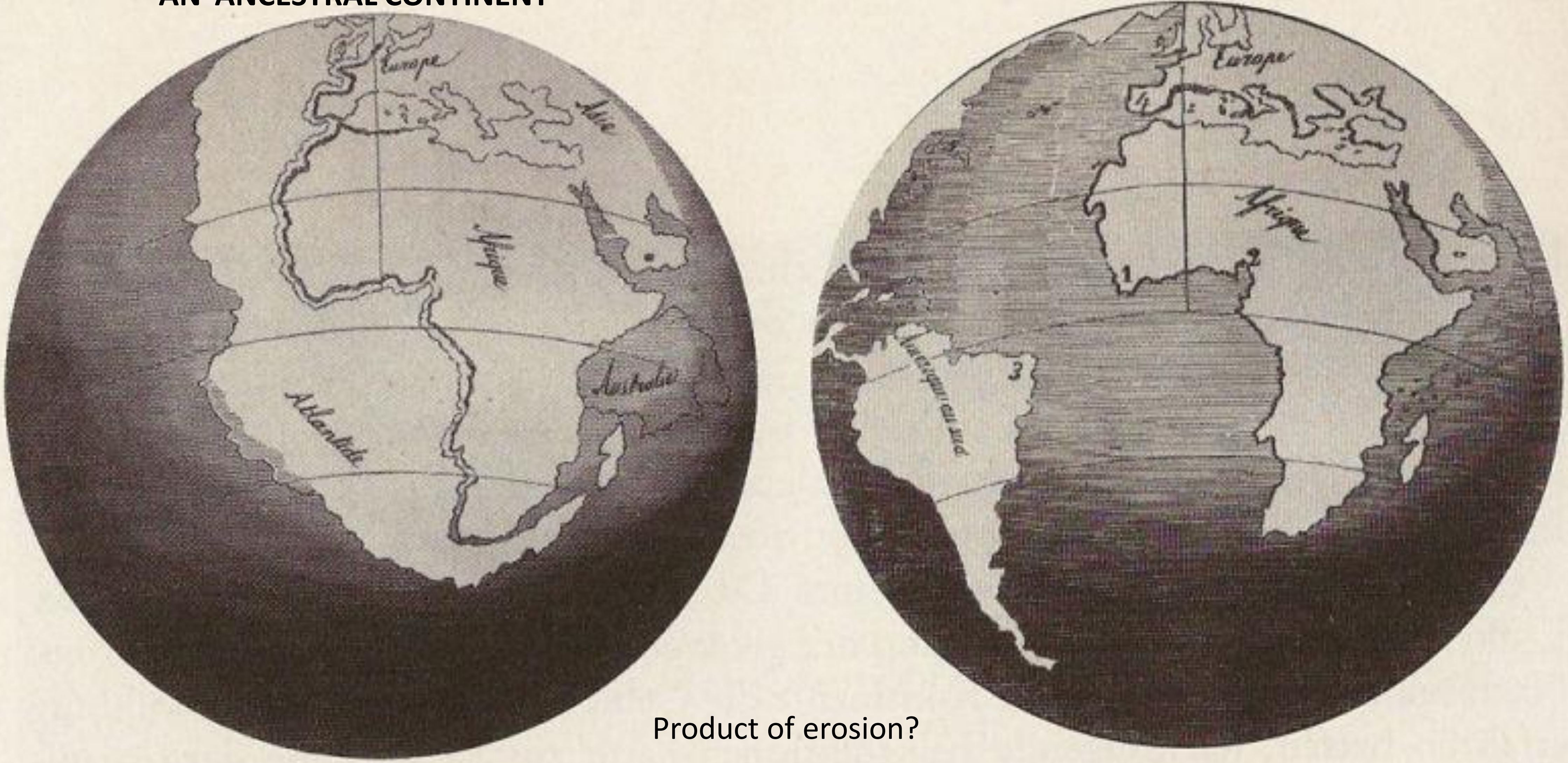
First World Atlas - 1570 (Abraham Ortelius)

Globalization -
World Trade -
Mercantilism -
Atlases -

- SIR FRANCIS BACON (and others): 1610 - Biblical Floods

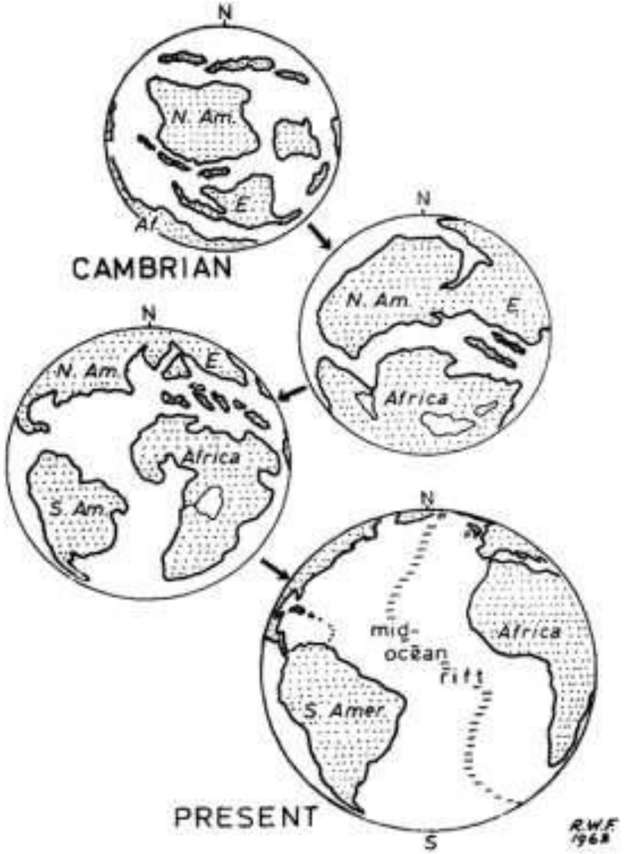
Scientific Proposal: ANTONIO SNIDER-PELLEGRINI 1858: same fossils in North America and Europe

AN 'ANCESTRAL CONTINENT'

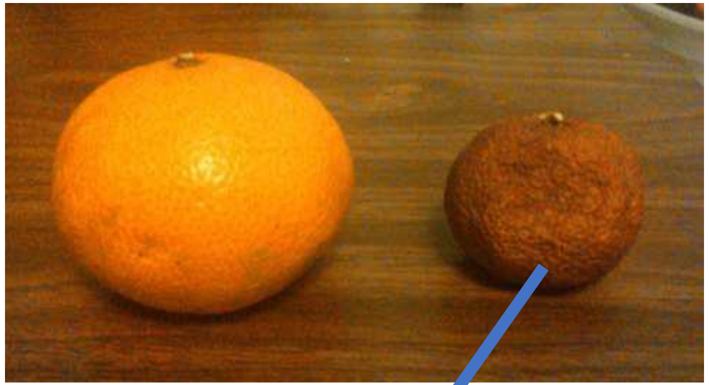


Product of erosion?

EXPANDING EARTH MODEL



CONTRACTING AND COOLING EARTH MODEL



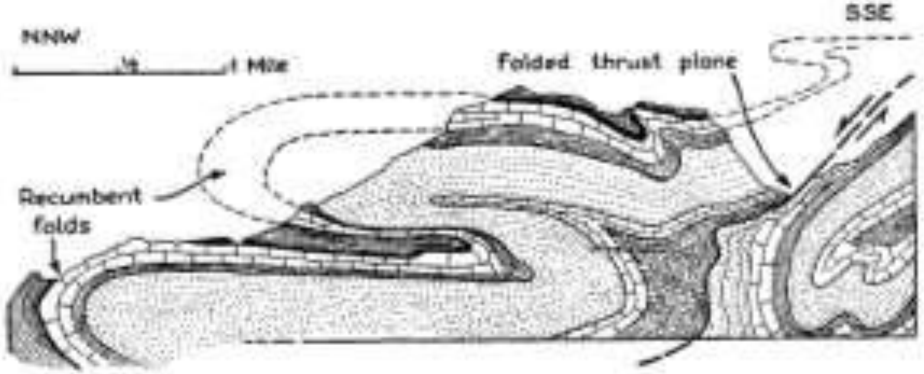


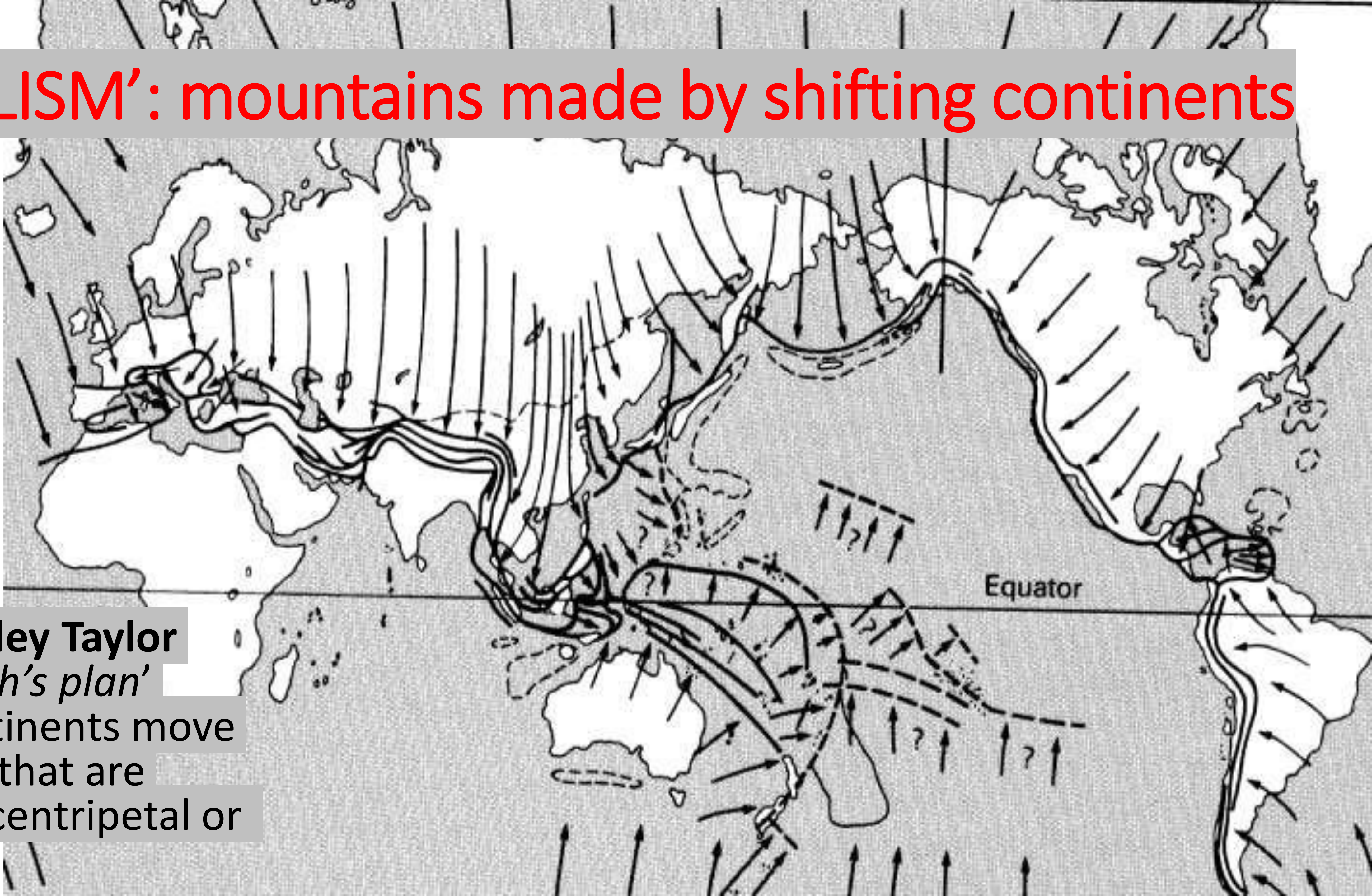
FIGURE 10-19. Recumbent folds and a folded thrust surface, Swiss Alps. (After A. Heim, 1922.)

‘PERMANENTISM’ - Continents are fixed in position connected by land bridges which sink to create oceans

- Carl Schuchert (1858-1942) Yale University

Still no continental movement!
Fixed continents joined by transient landbridges

‘MOBILISM’: mountains made by shifting continents



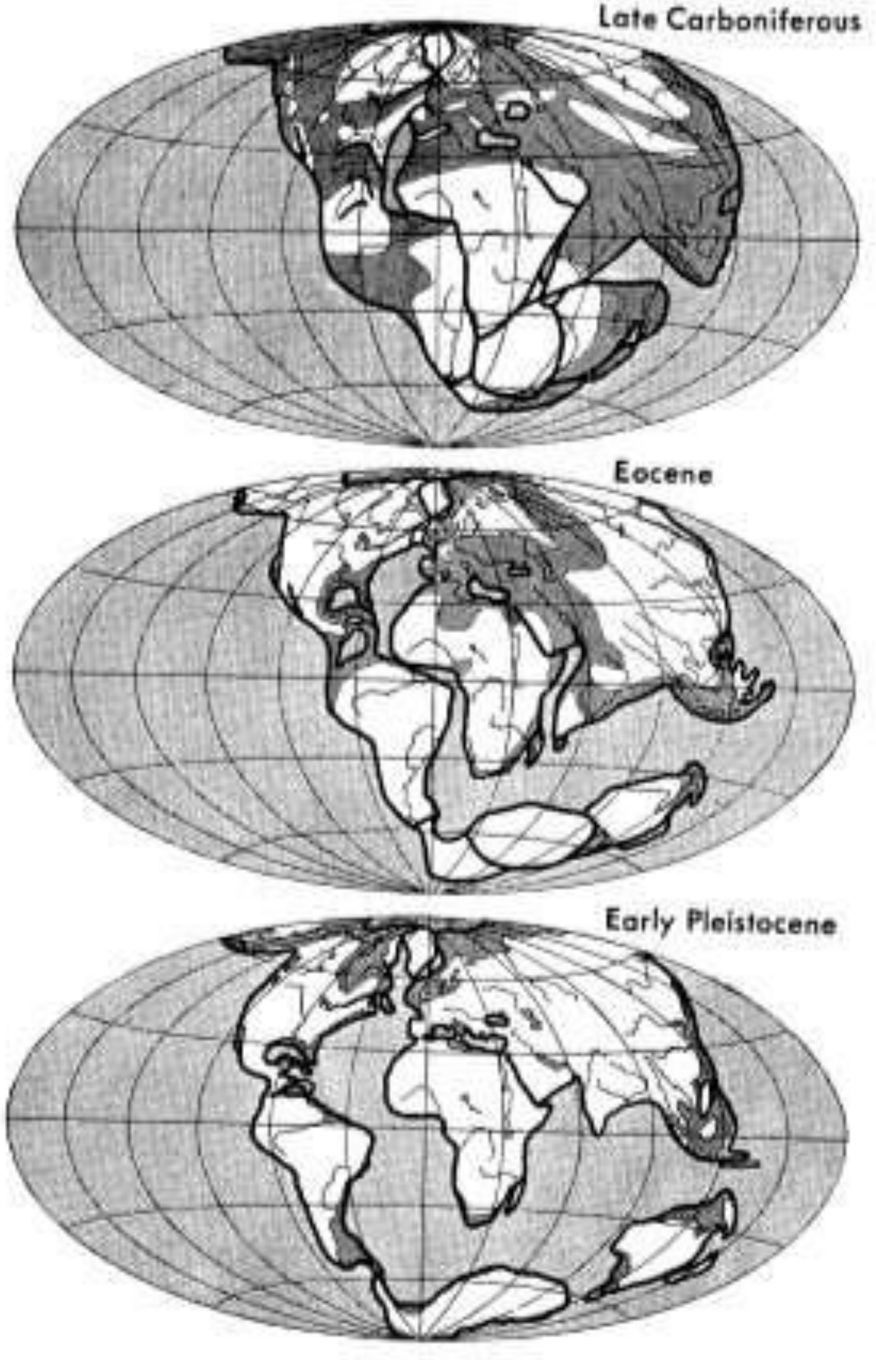
Frank Bursley Taylor
1910: ‘*Earth’s plan*’ where continents move as ‘sheets’ that are moved by centripetal or tidal forces

Alfred Wegener:

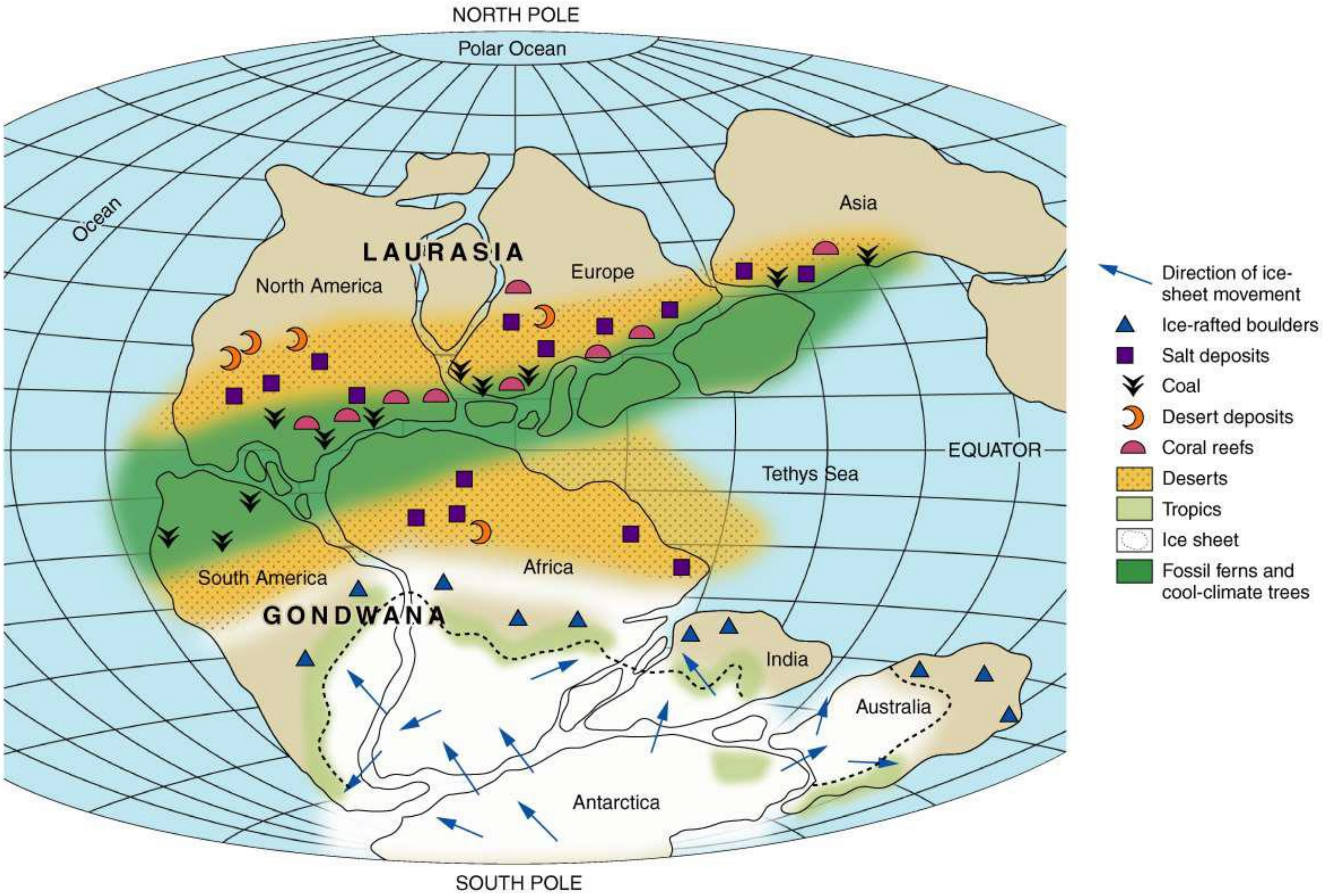
***1915 -** Die Entstehung der Kontinen*
und Ozeane (“The Origin of Continents and Oceans”).

Made models, plotted fossils and geology

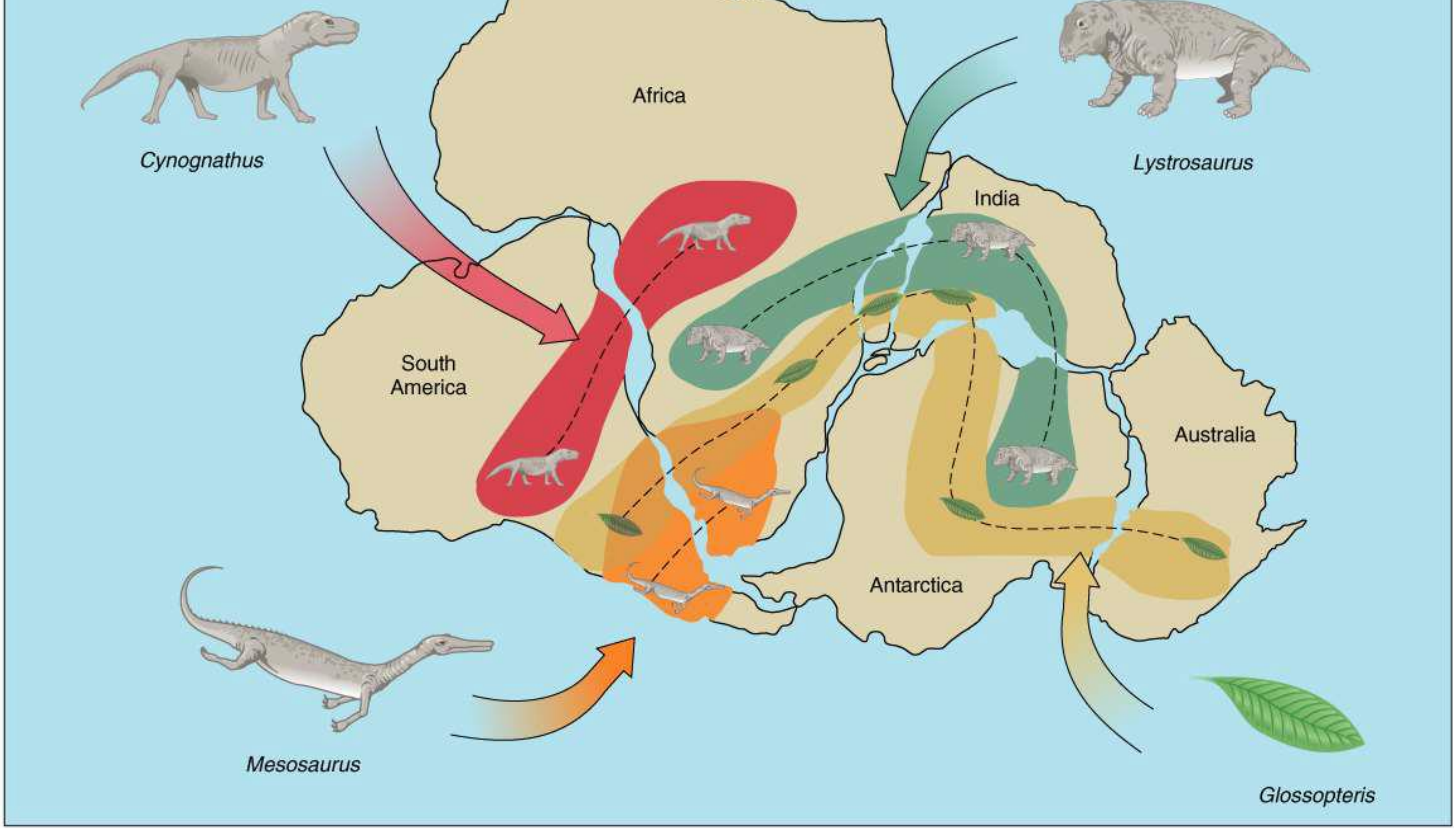
PANGEA (‘ALL THE LANDS’)



Matching rock types of the same age from one continent to another to reconstruct Pangea



And it made sense of the fossil record on continents that are now far apart

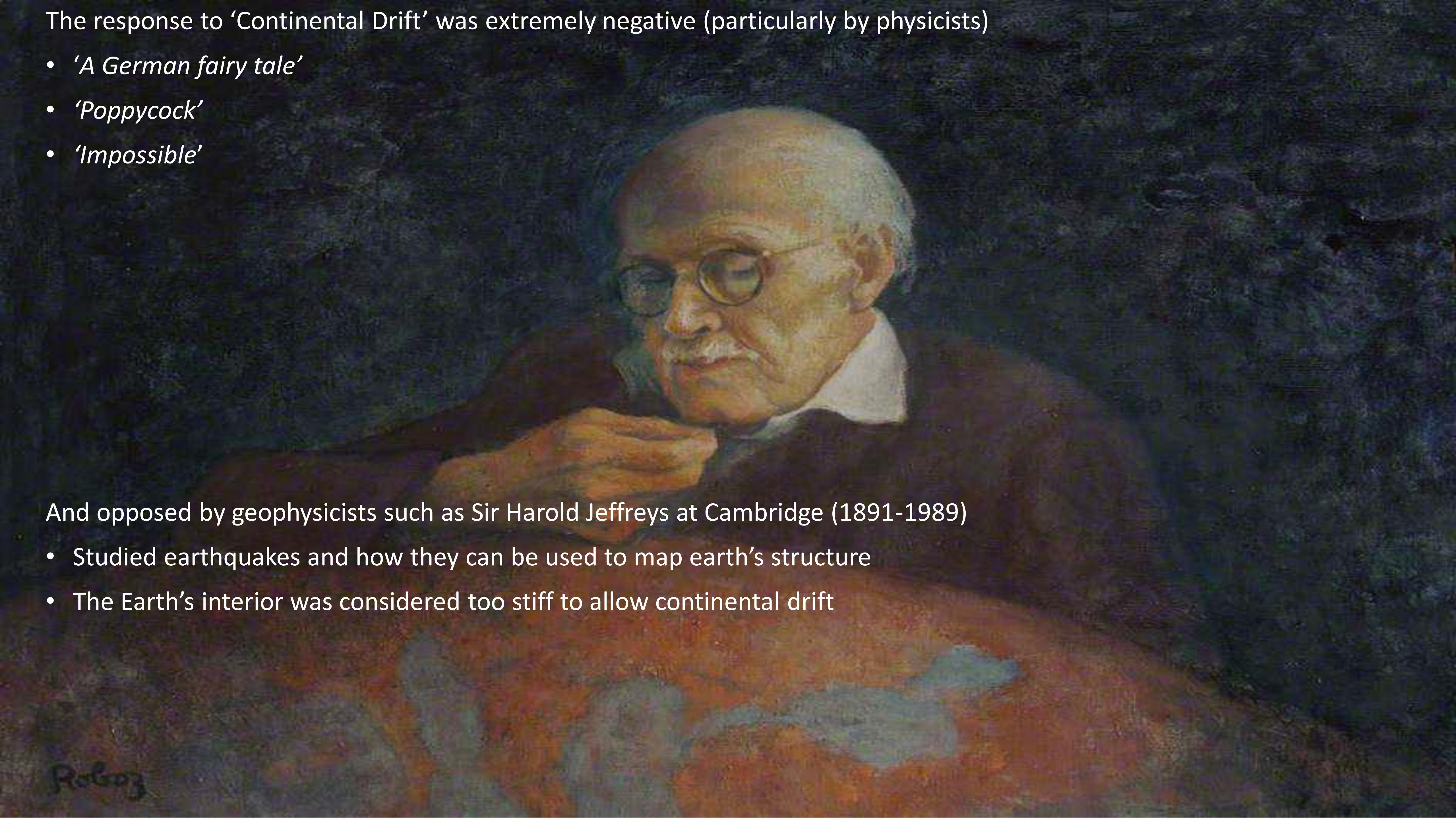


The response to ‘Continental Drift’ was extremely negative (particularly by physicists)

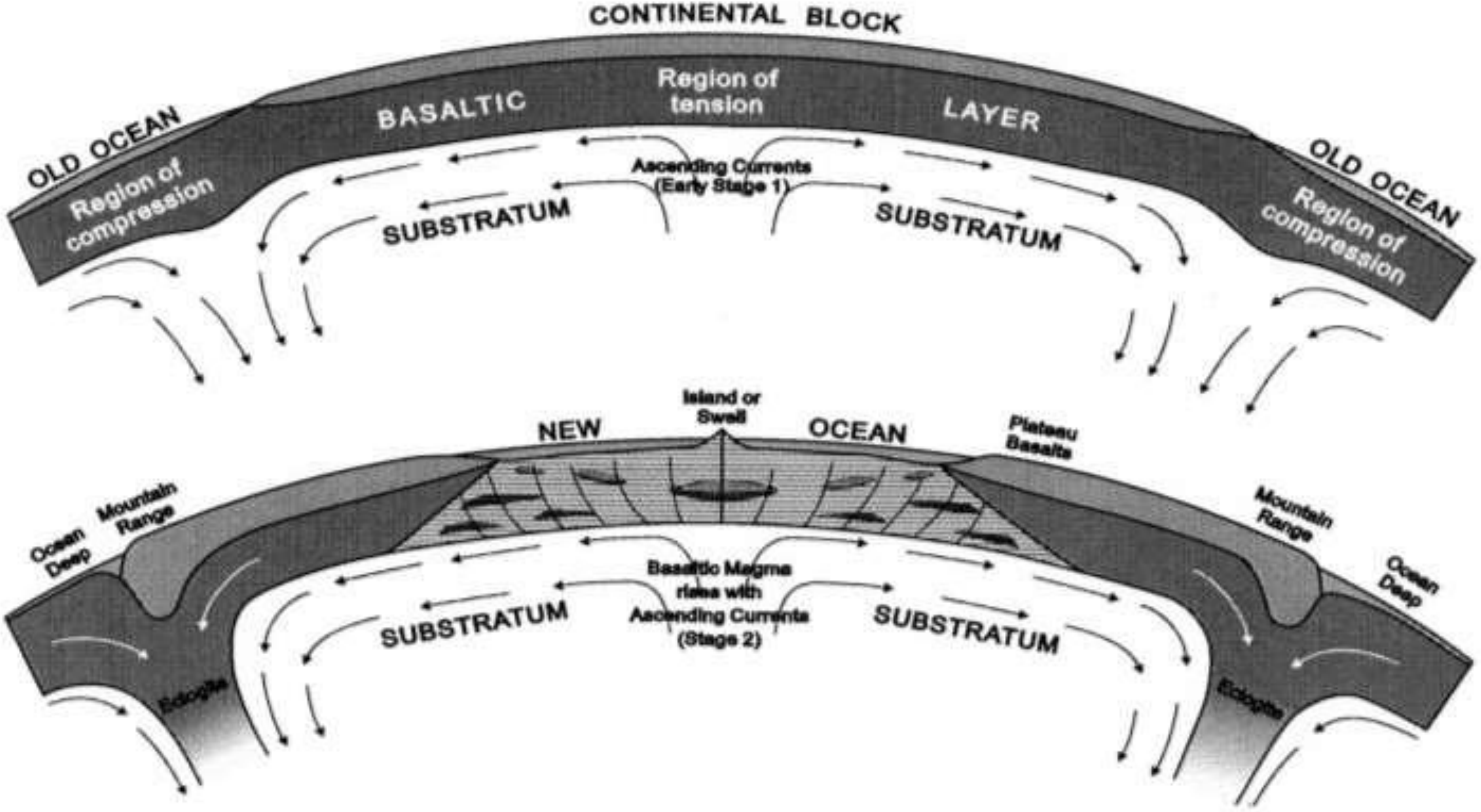
- *‘A German fairy tale’*
- *‘Poppycock’*
- *‘Impossible’*

And opposed by geophysicists such as Sir Harold Jeffreys at Cambridge (1891-1989)

- Studied earthquakes and how they can be used to map earth’s structure
- The Earth’s interior was considered too stiff to allow continental drift



Mantle Convection

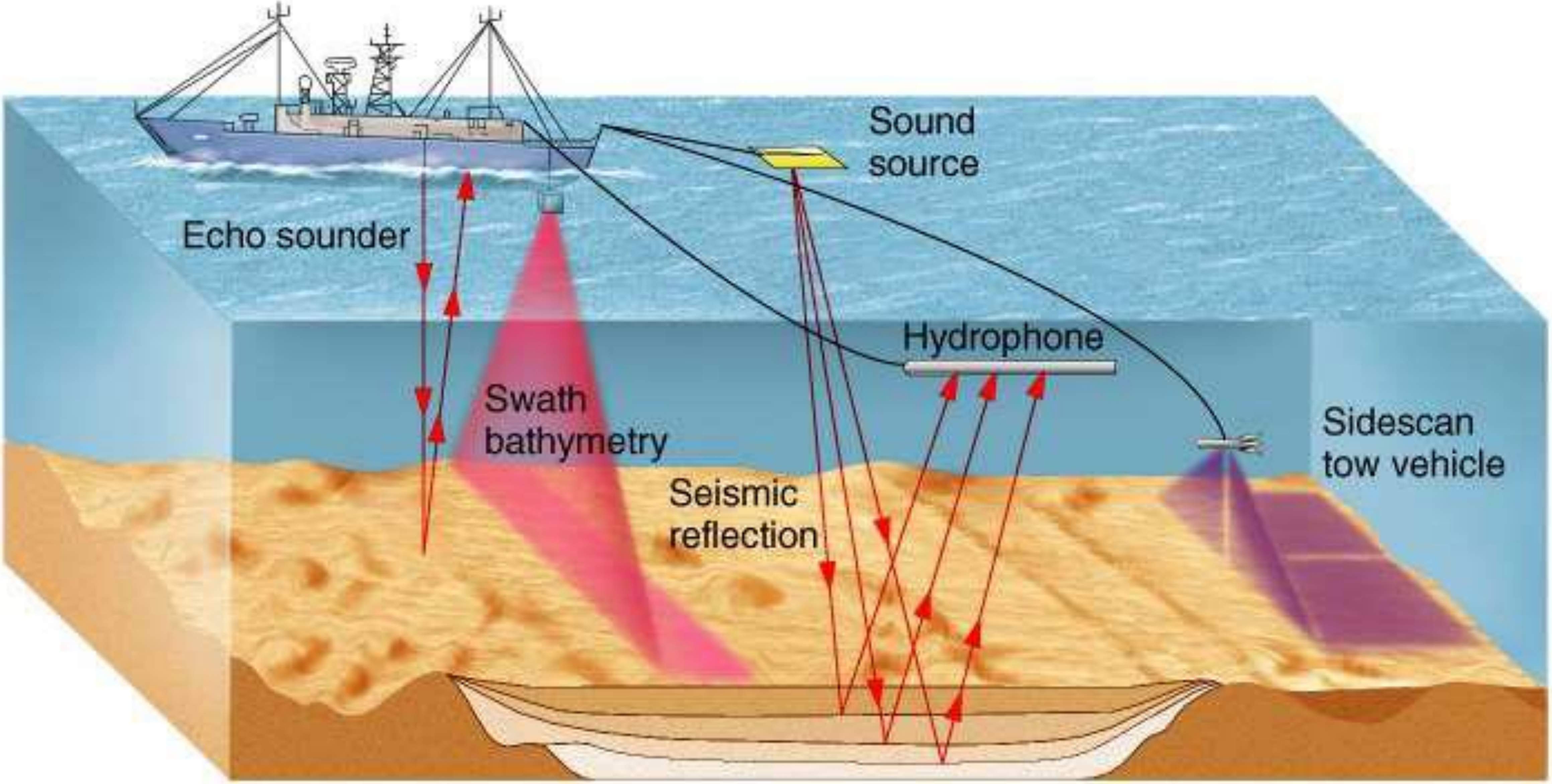


Arthur Holmes (1928) Convection currents driving the drift of continents
-largely ignored in North America, had trouble getting **published**

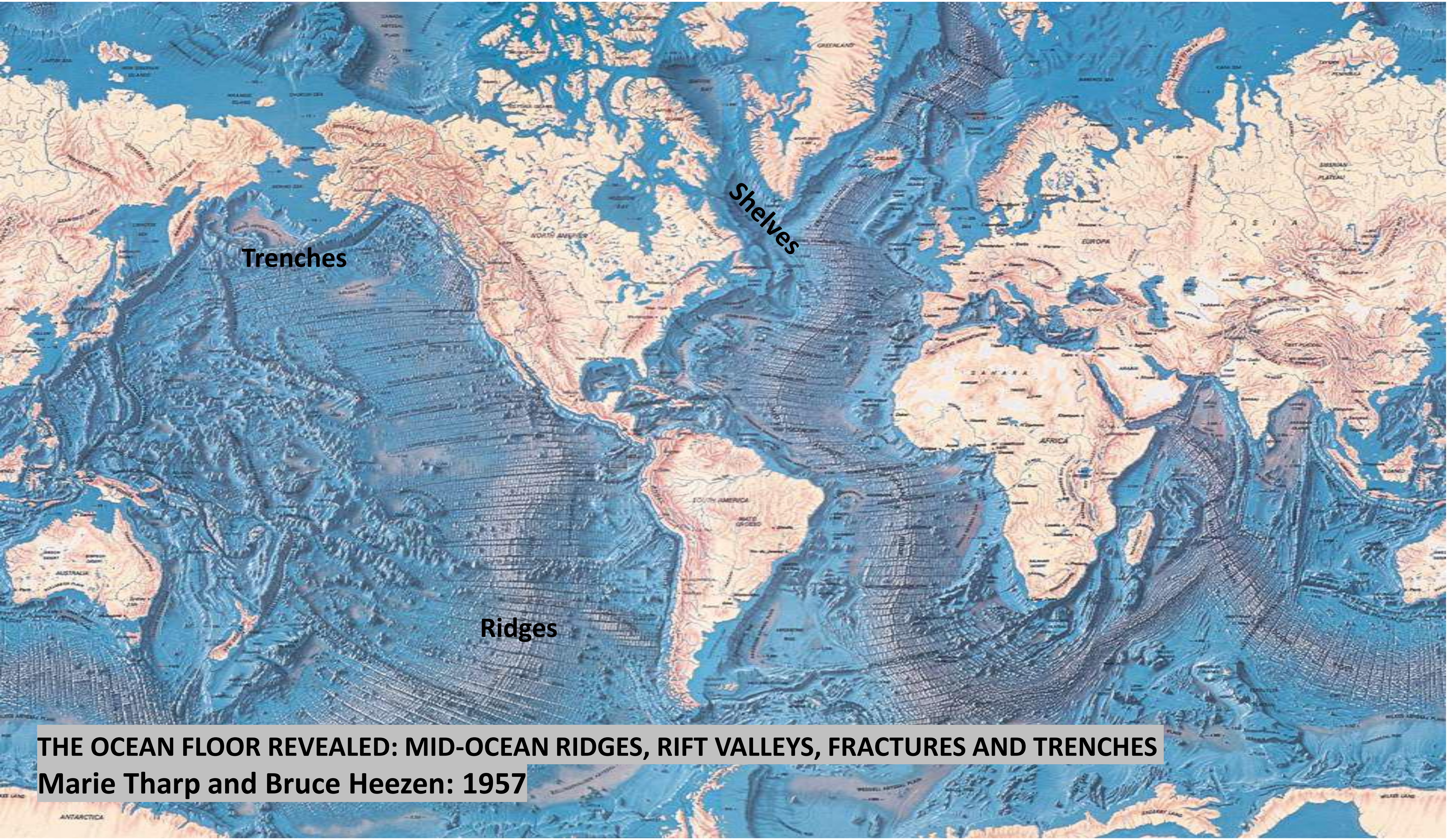
Data Revolution!

1939-1945 Second World War
followed by the Cold War beginning in 1948 resulted in anti-submarine warfare and the development of new sonar devices to find them and map the ocean floor

1960's: THE OCEAN FLOOR GETS MAPPED (Geophysics)

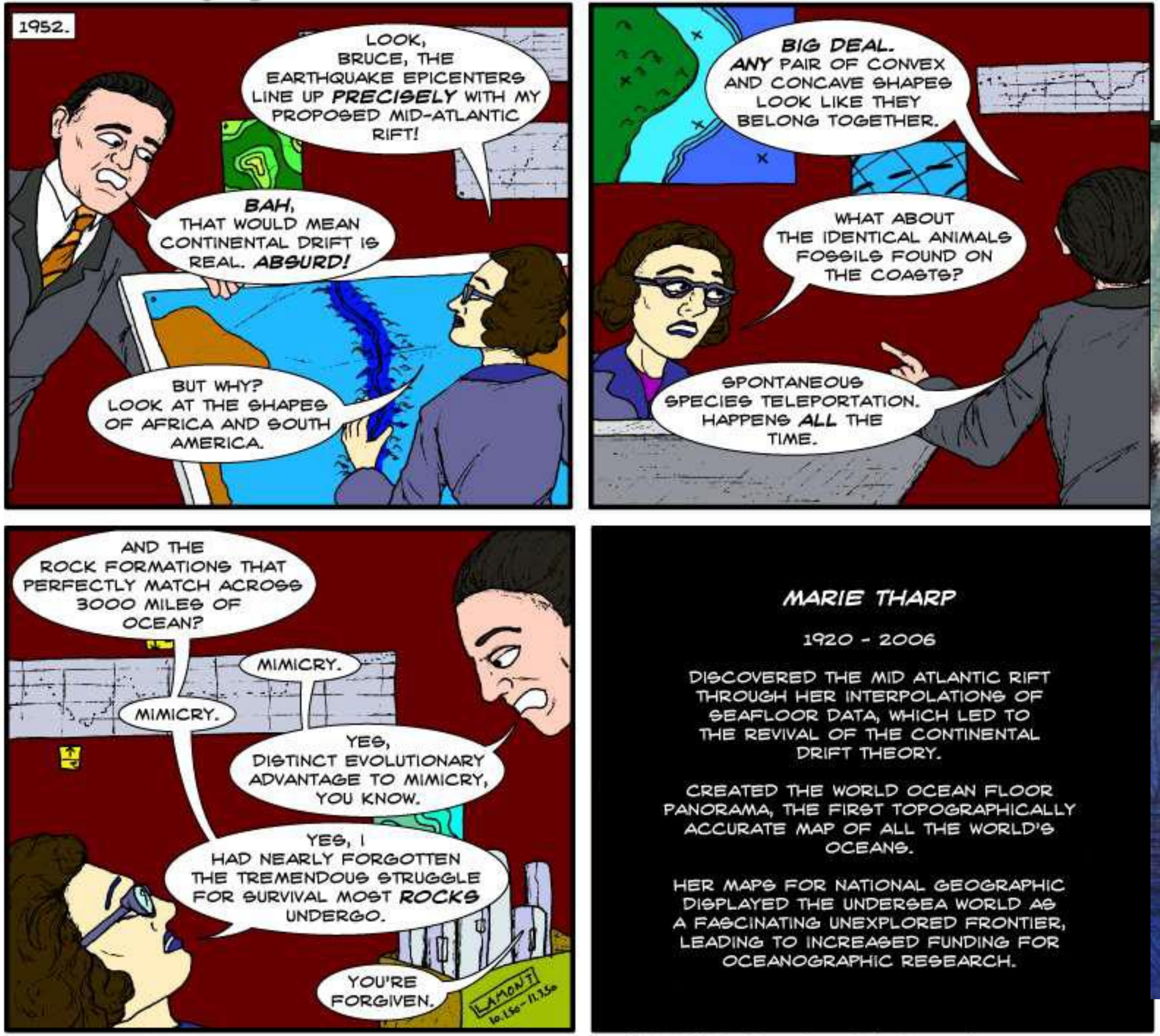


Magnetometer

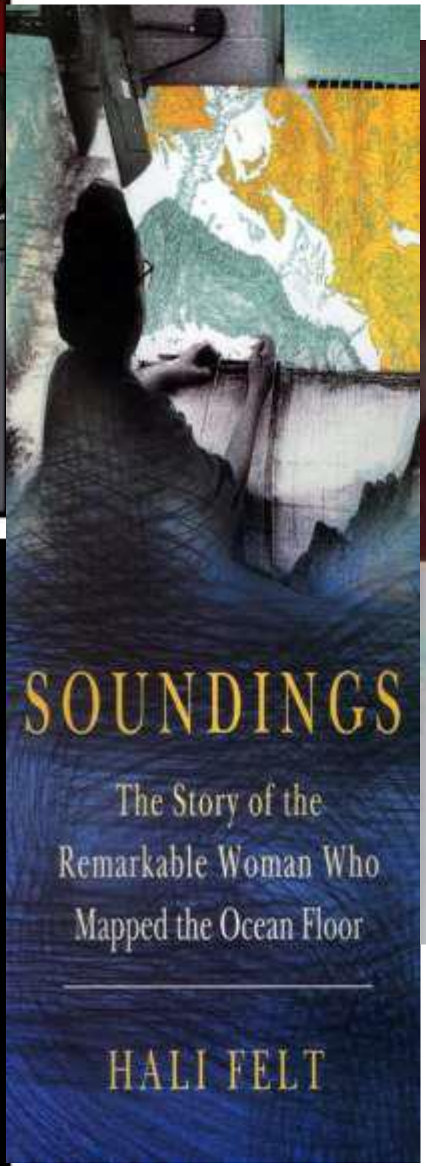


THE OCEAN FLOOR REVEALED: MID-OCEAN RIDGES, RIFT VALLEYS, FRACTURES AND TRENCHES
Marie Tharp and Bruce Heezen: 1957

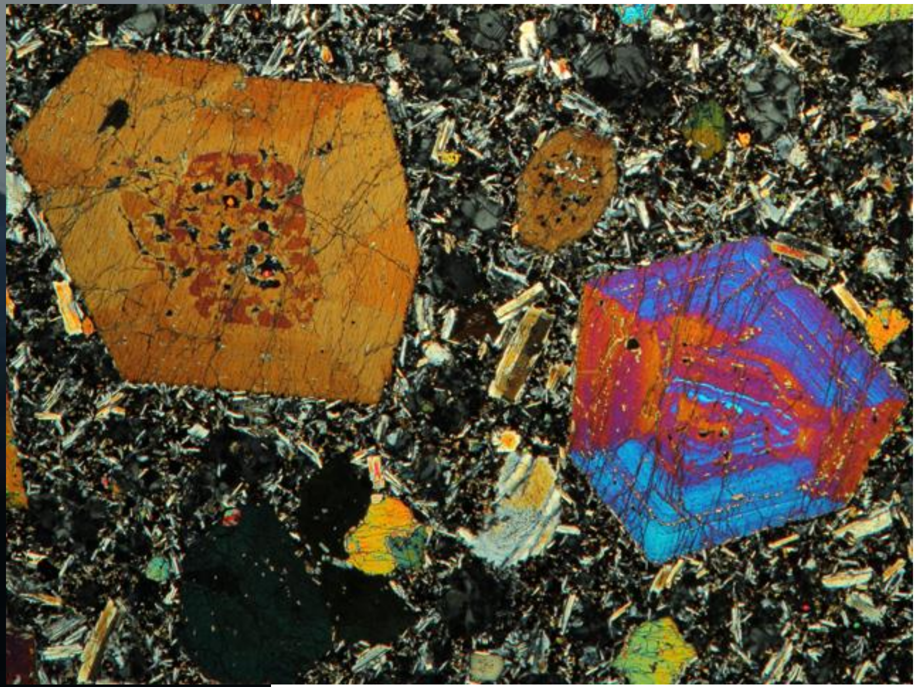
The Oceanographic Dress Code: No Bare Mid Rifts



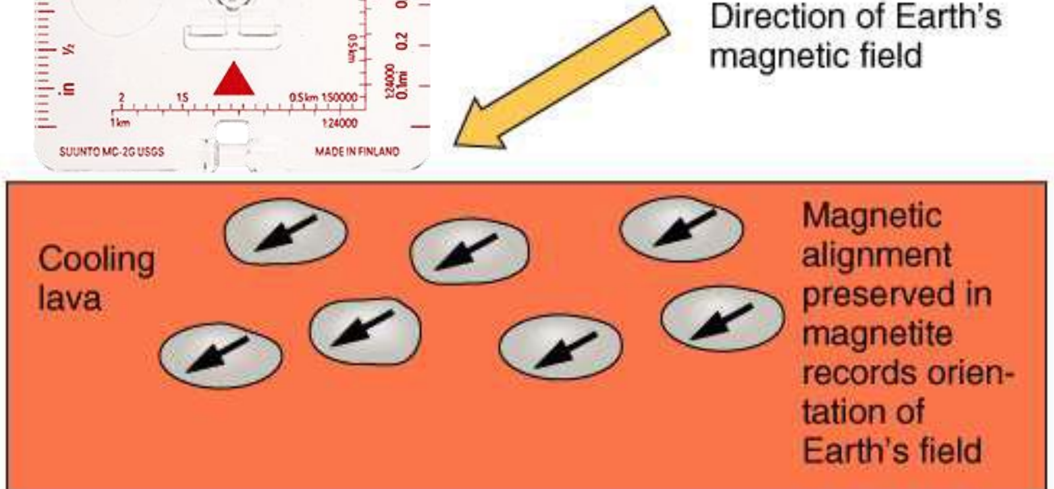
www.womenyoushouldknow.net/category/women-in-science/



Magma – molten fluid rock. A mix of fluid and crystals

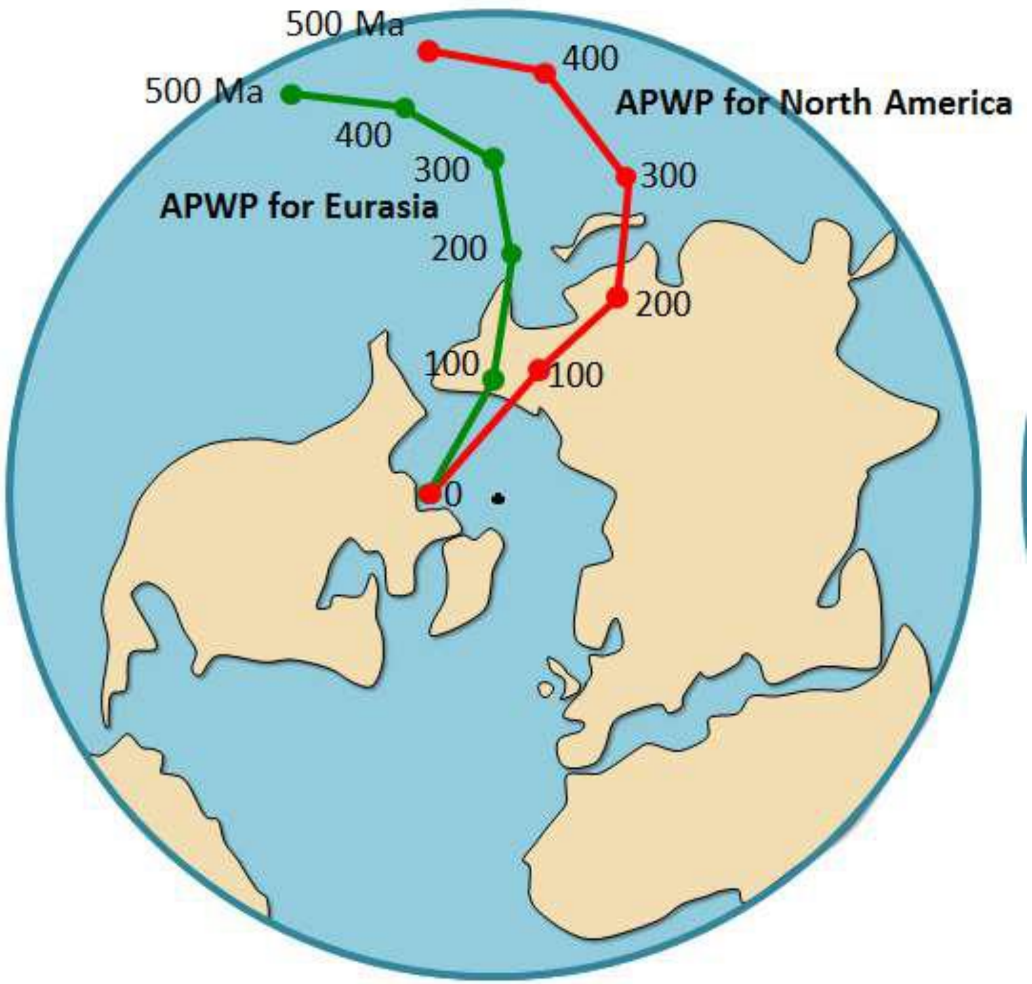


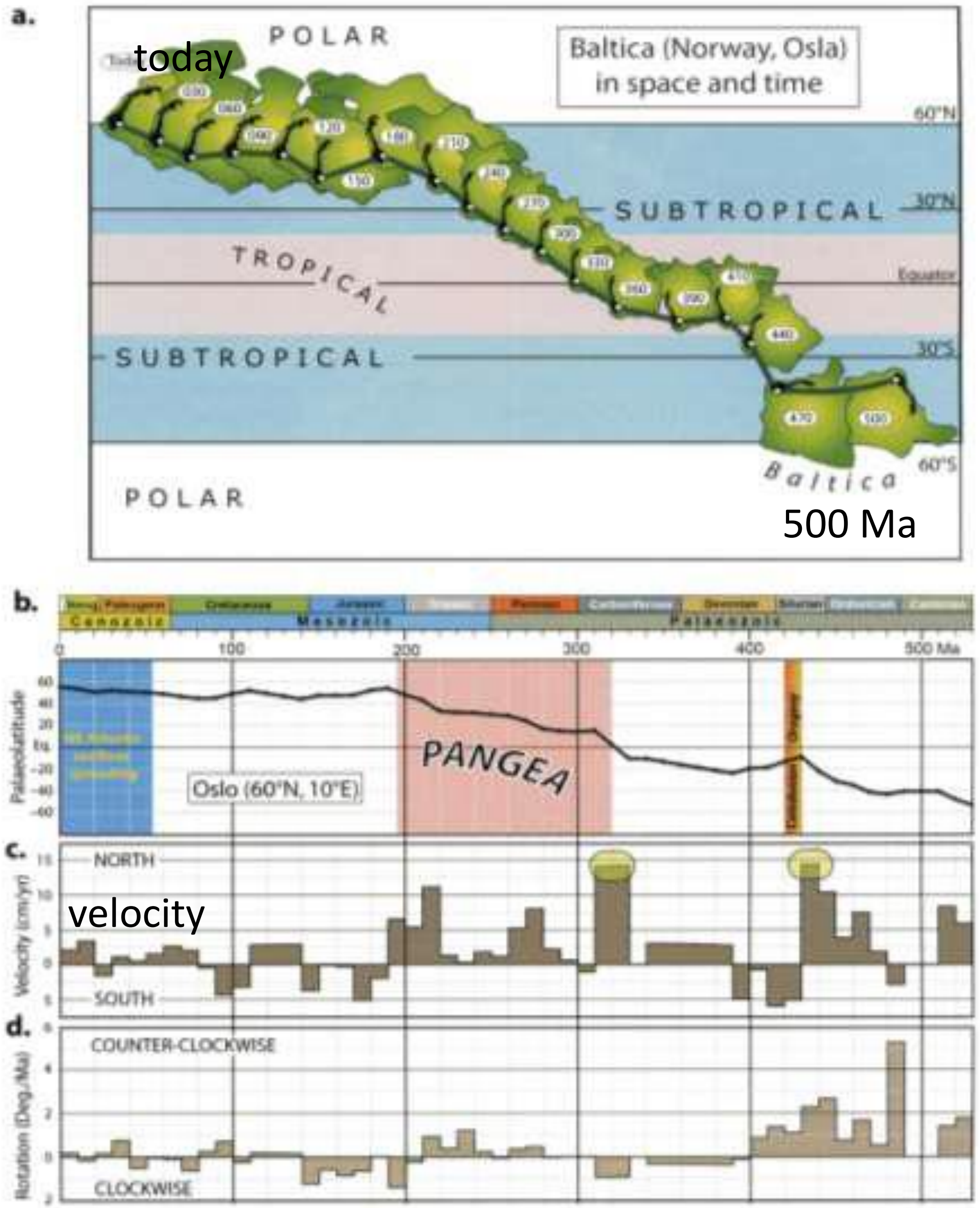

SUUNTO
SUUNTO MC-2G USGS
MADE IN FINLAND


Direction of Earth's magnetic field
Cooling lava
Magnetic alignment preserved in magnetite records orien-tation of Earth's field

LATE 1950'S:

- Magnetic grains in old rocks don't point north!
- APPARENT POLAR WANDER CURVES
- Ted Irving, Trond Torsvik and others



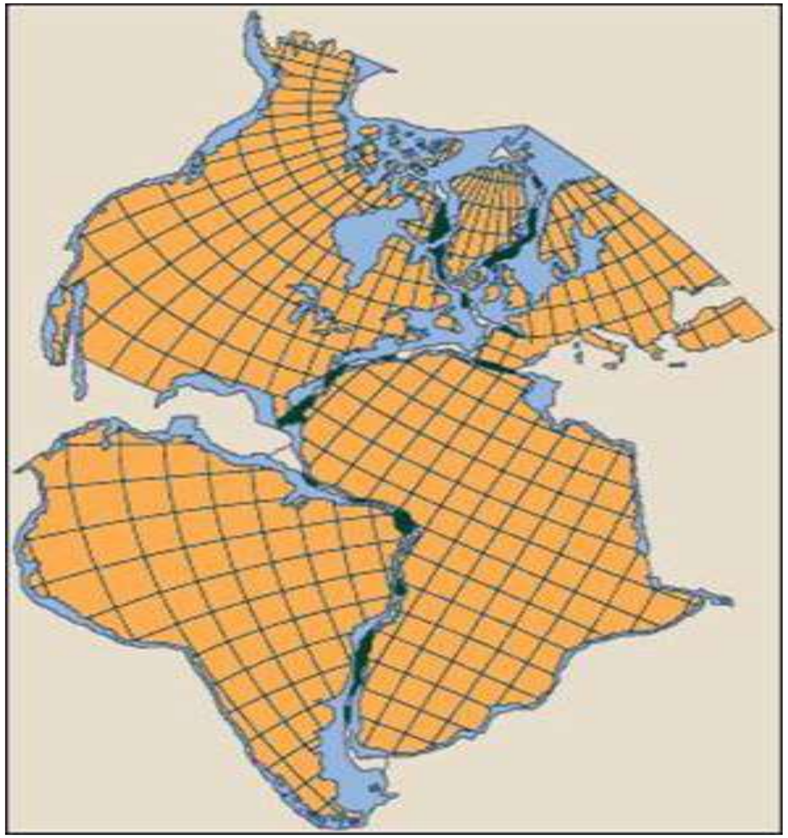


Confirms that continents do drift and at which rates

1960's – Computers in universities

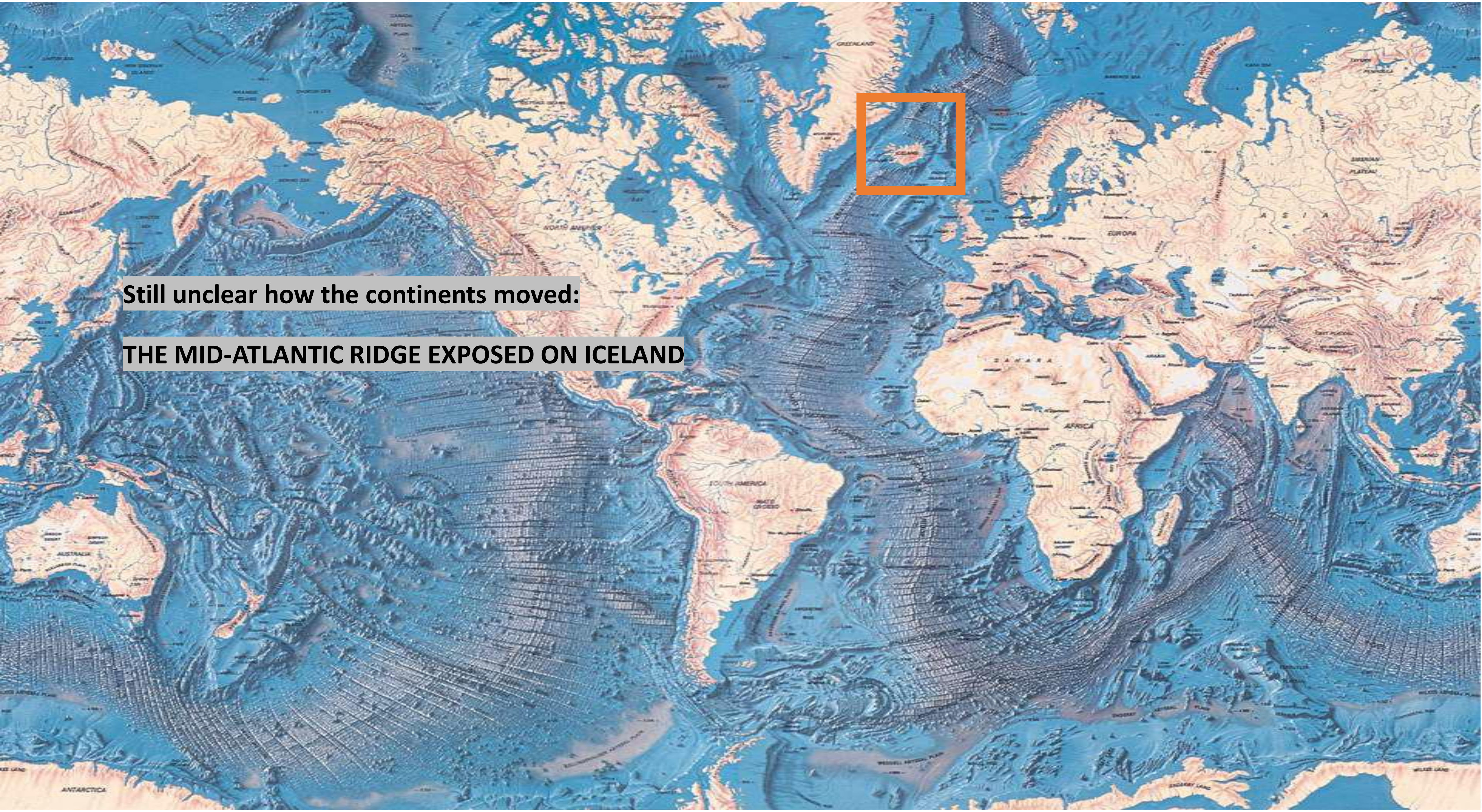
Edward Bullard, 1965:

- **Computer reconstruction confirmed the 'fit' of modern day continents either side of the Atlantic Ocean**



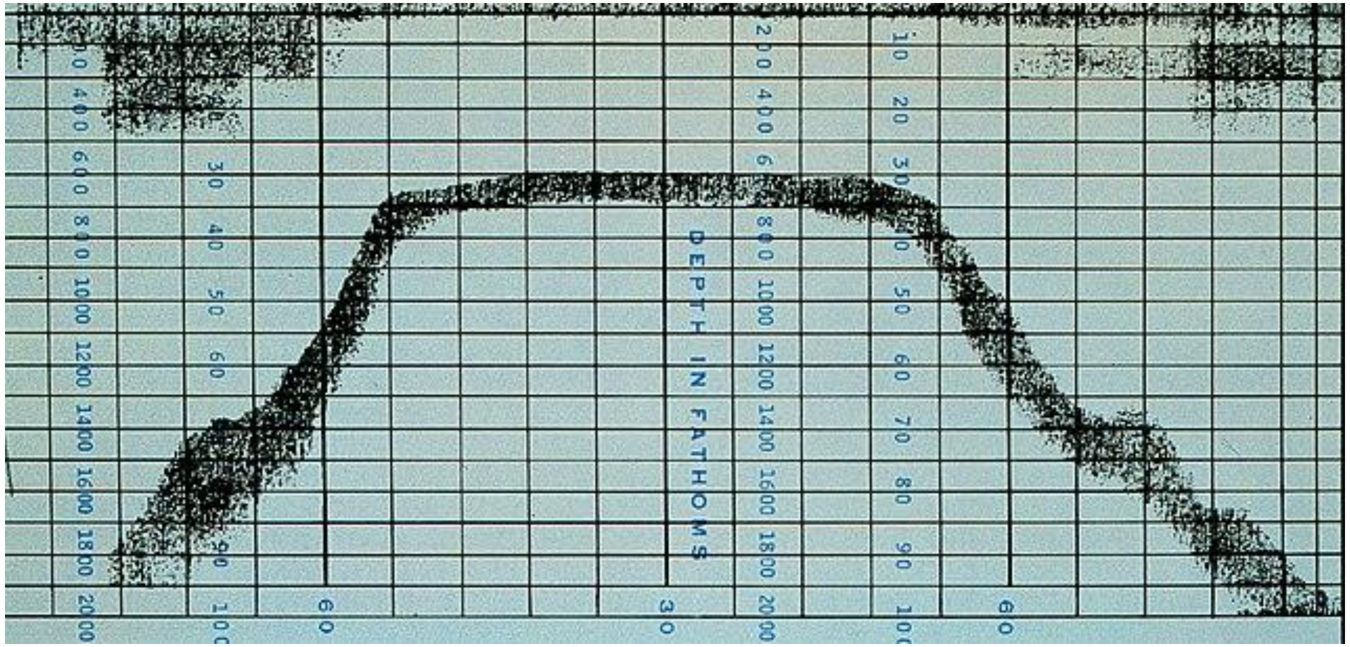
Still unclear how the continents moved:

THE MID-ATLANTIC RIDGE EXPOSED ON ICELAND



Harold Hess - Sea Floor Spreading

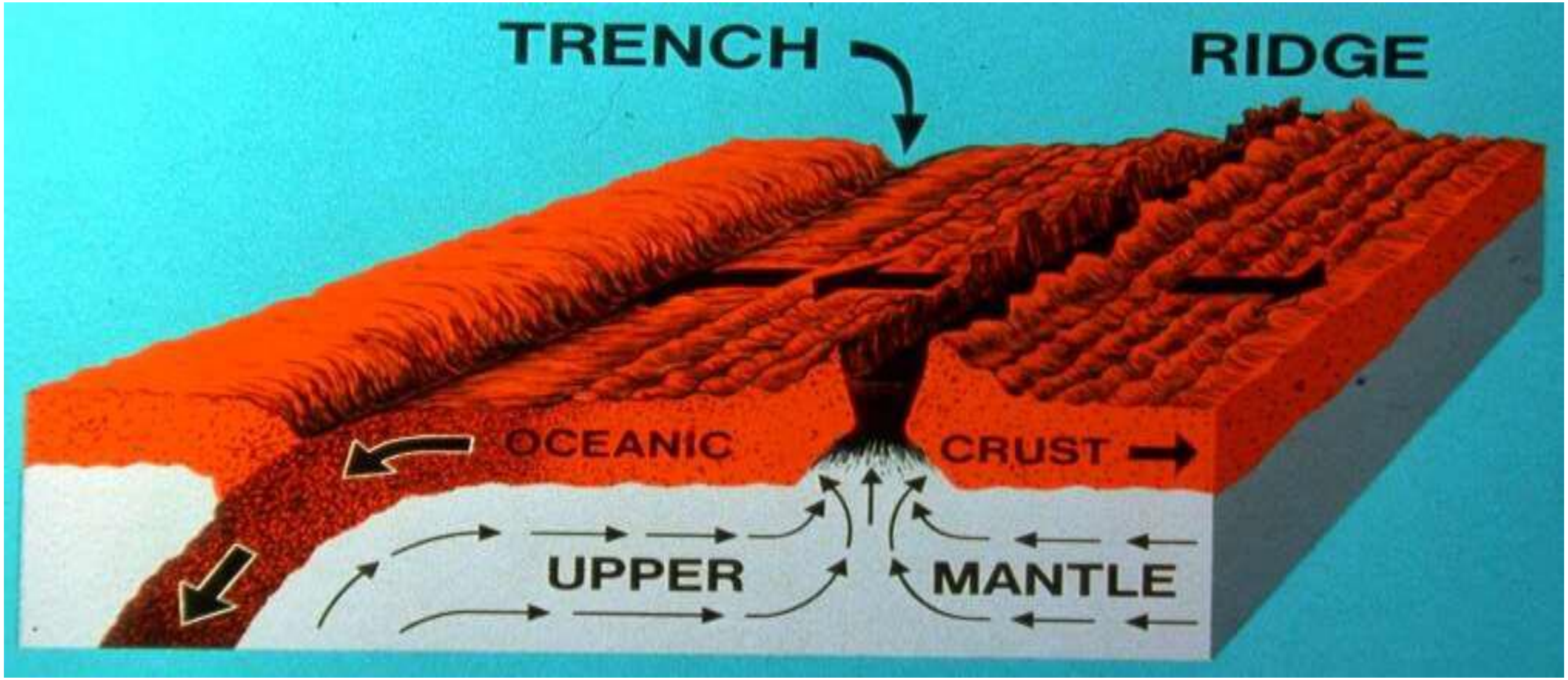
Guyots (sea mounts)



“Geopoetry”

Harold Hess introduced idea of ***sea floor spreading*** (1962)

- Sea floor created at MOR’s recycled into mantle at trenches
- Implied continents are moving!



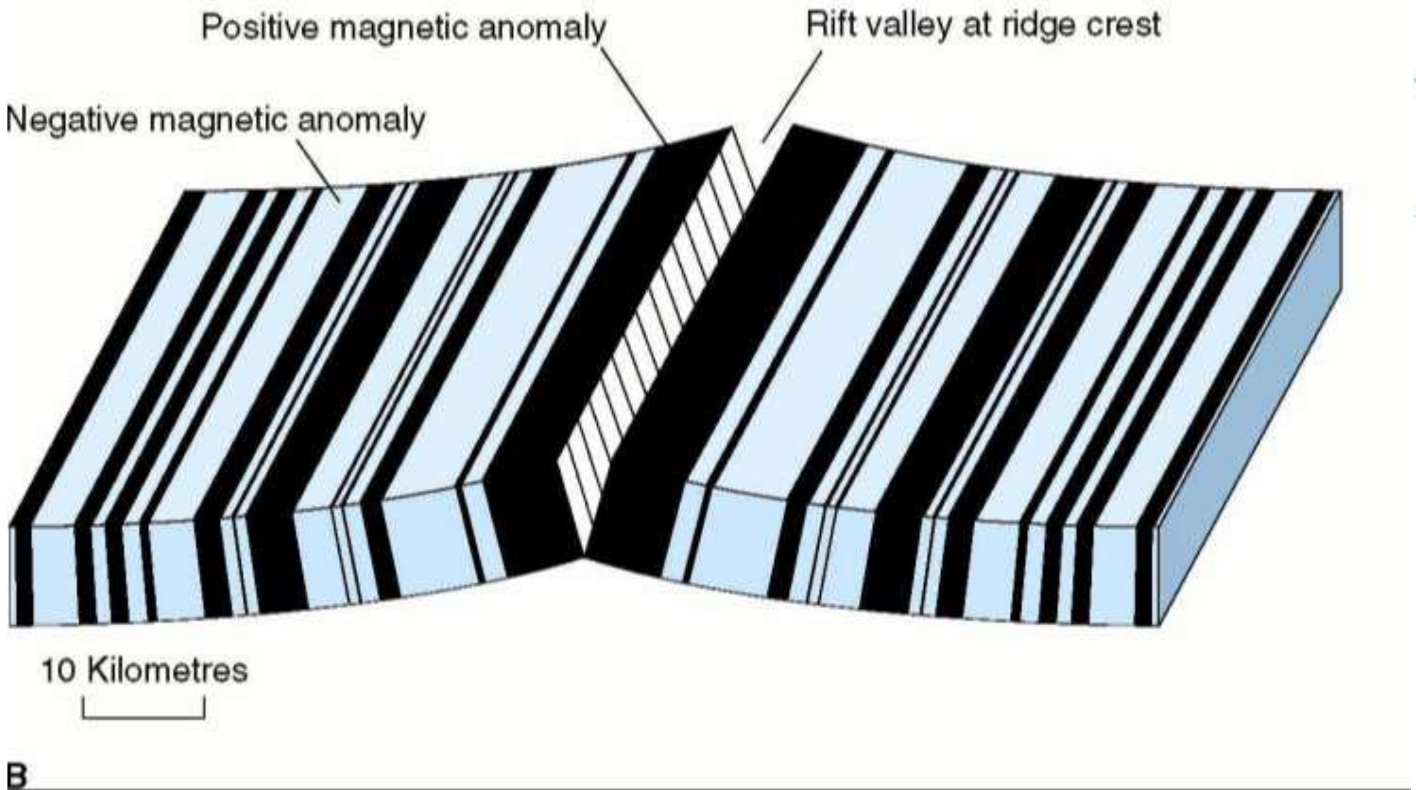
Subduction:
Plates are destroyed

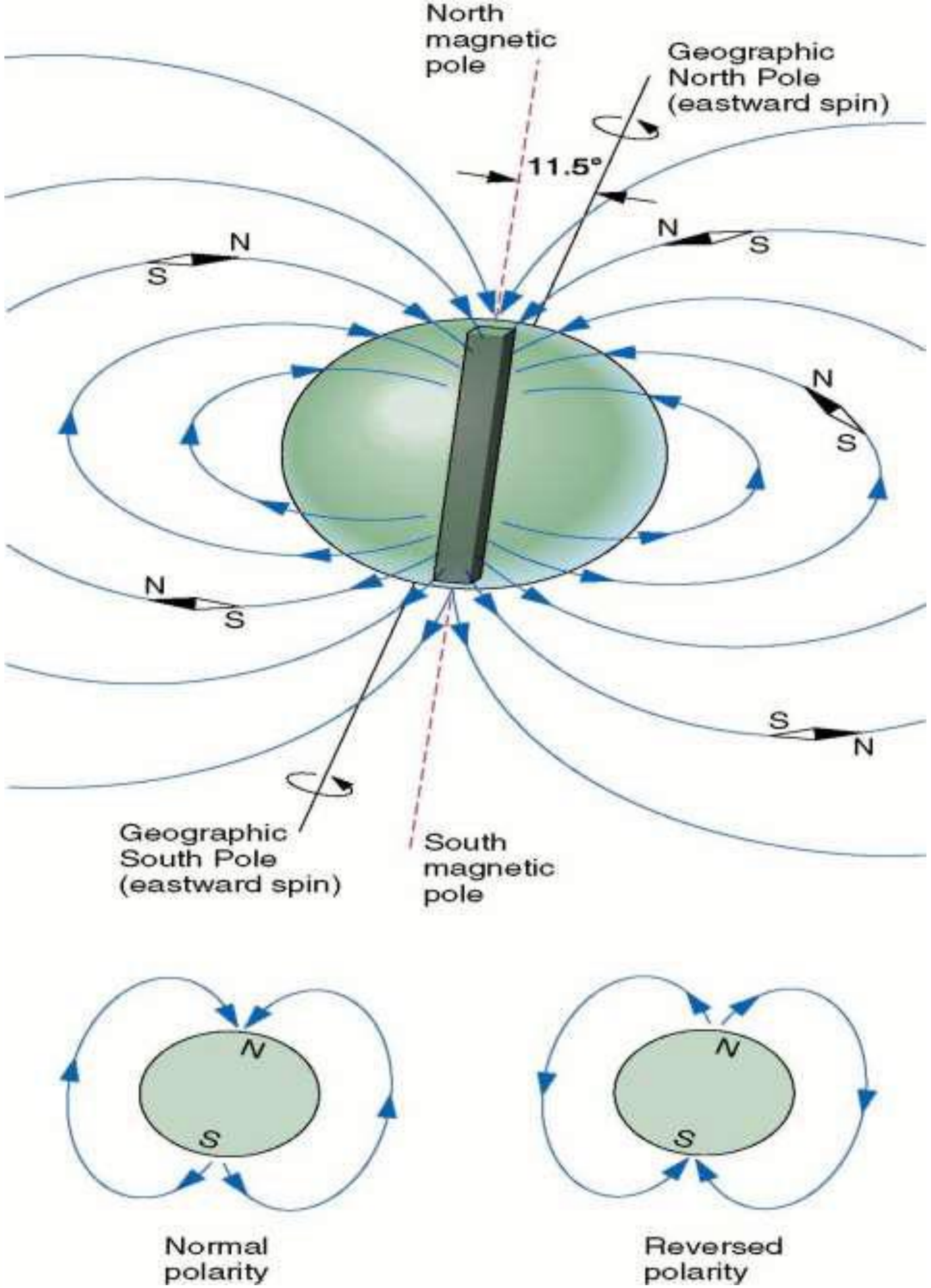
Mid Ocean Ridges:
Plates created

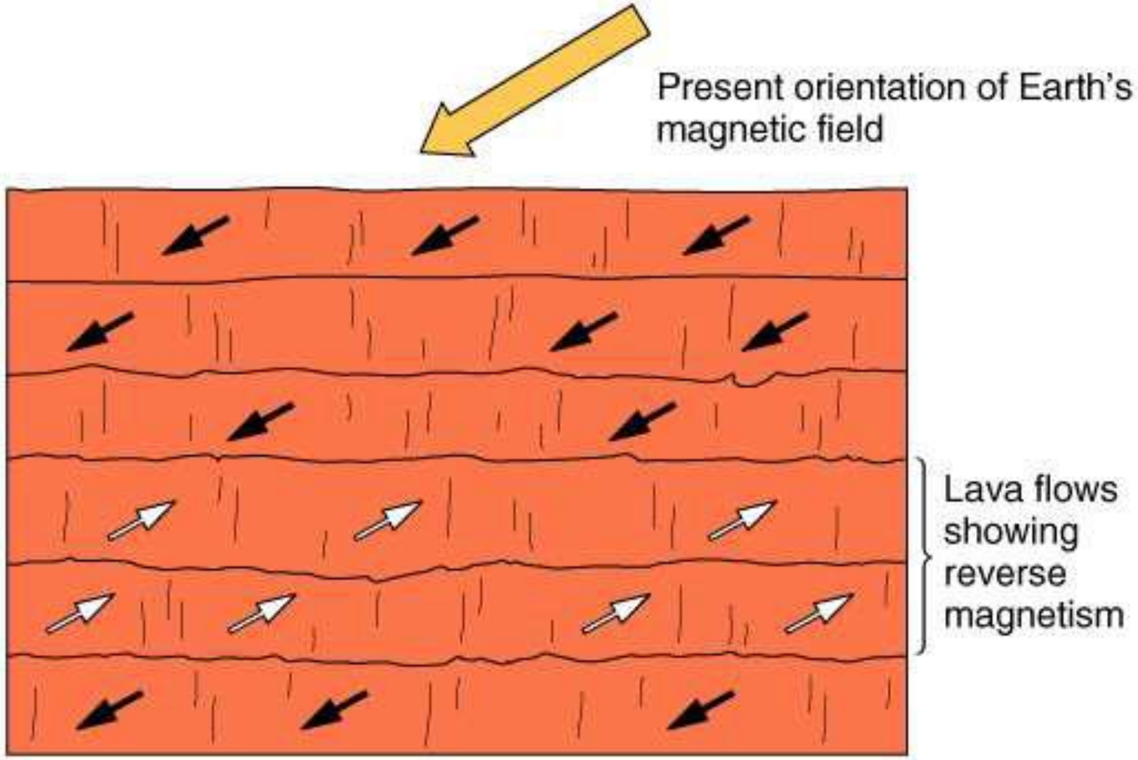
How do we test?

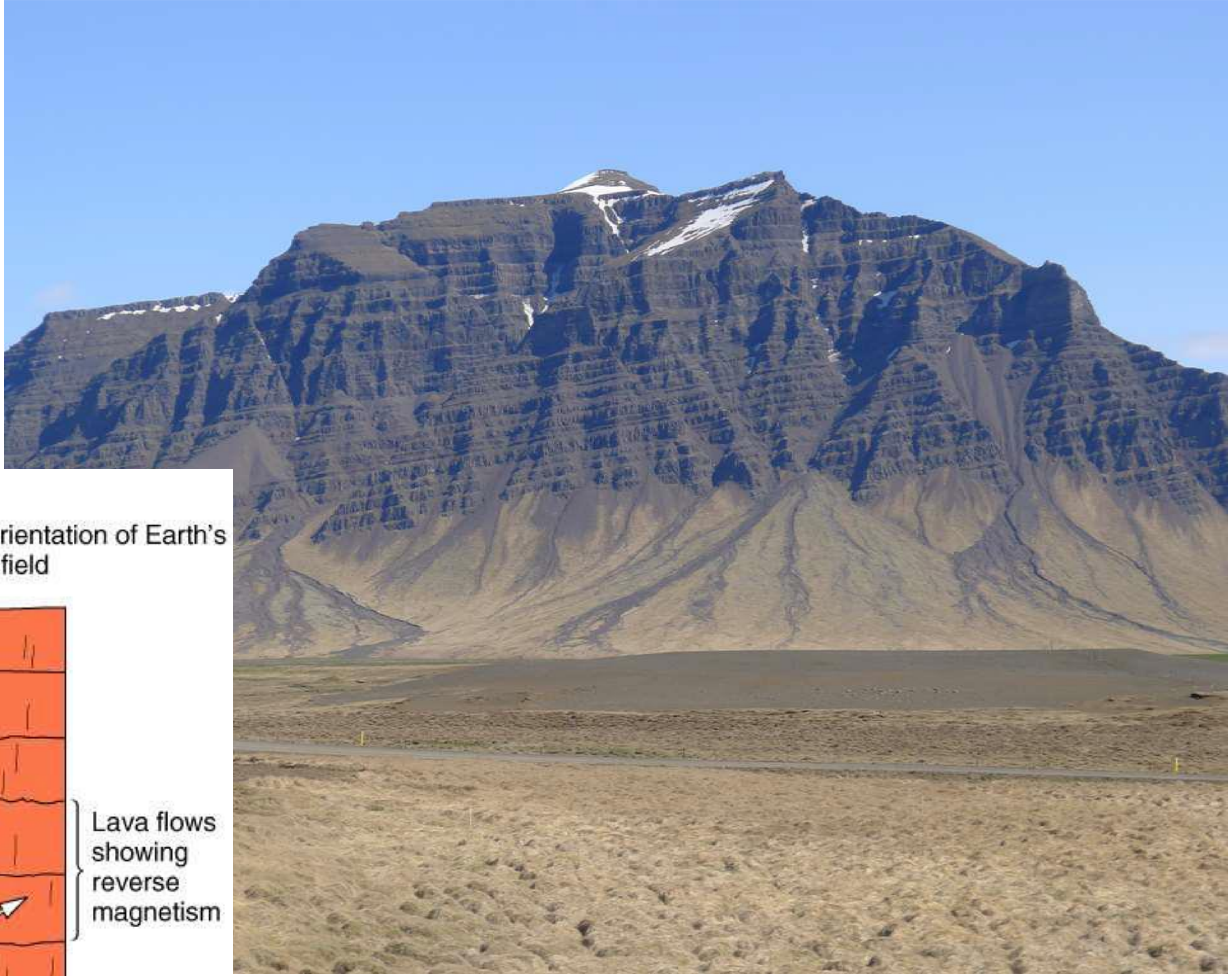
MAGNETIC STRIPES (1965)
The Vine-Mathews-Morely hypothesis

Magnetic Reversals: the ‘polarity’ of the magnetic field switches from N to S.
Magnetometer surveys of ocean floor identified patterns of anomalies on *each side of MORs*





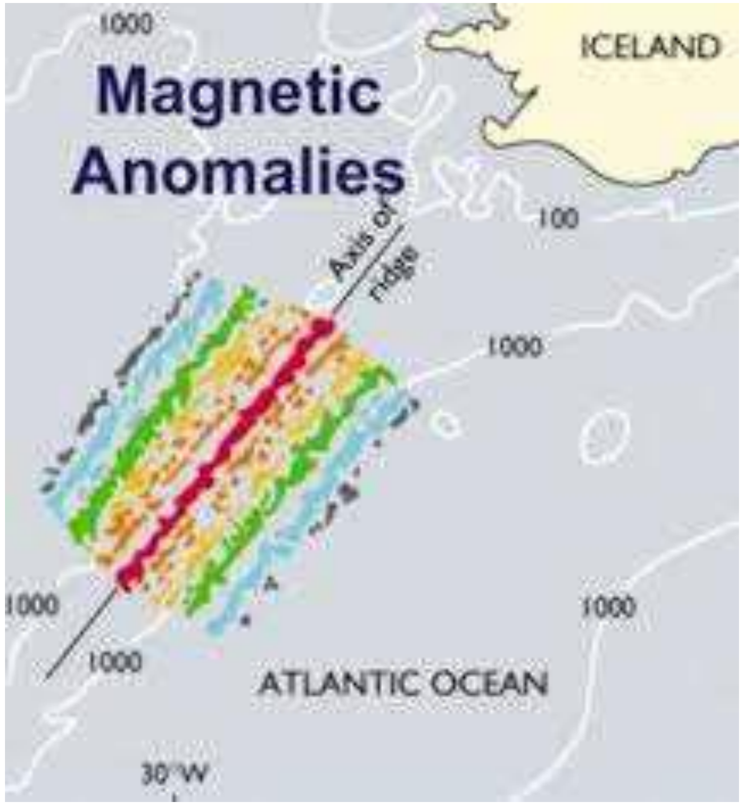

Present orientation of Earth's magnetic field
Lava flows showing reverse magnetism

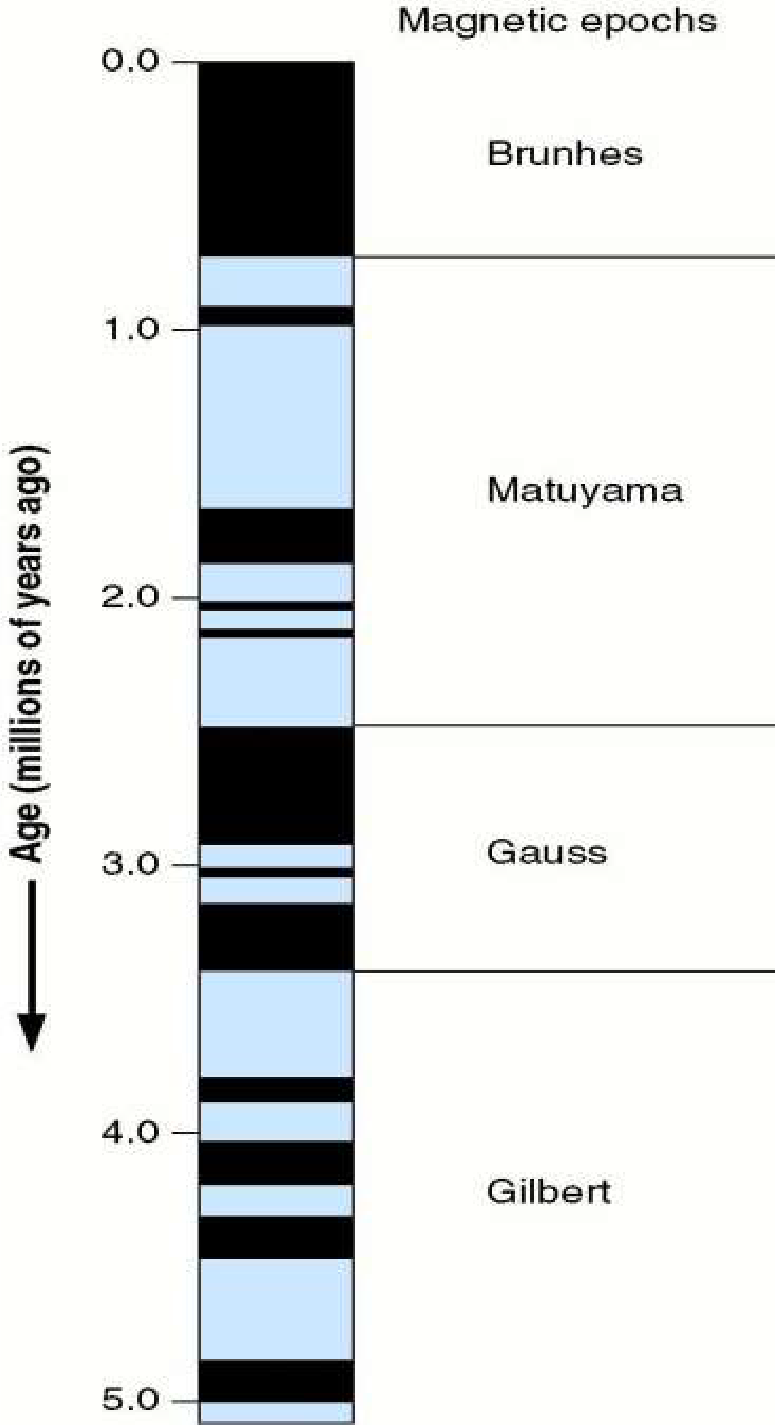


Can date reversals

Paleomagnetic polarity changes through time

- Magnetic reversals occur on average every 500,000 years (with variation)
- Takes about 10,000 years or less for a reversal to occur

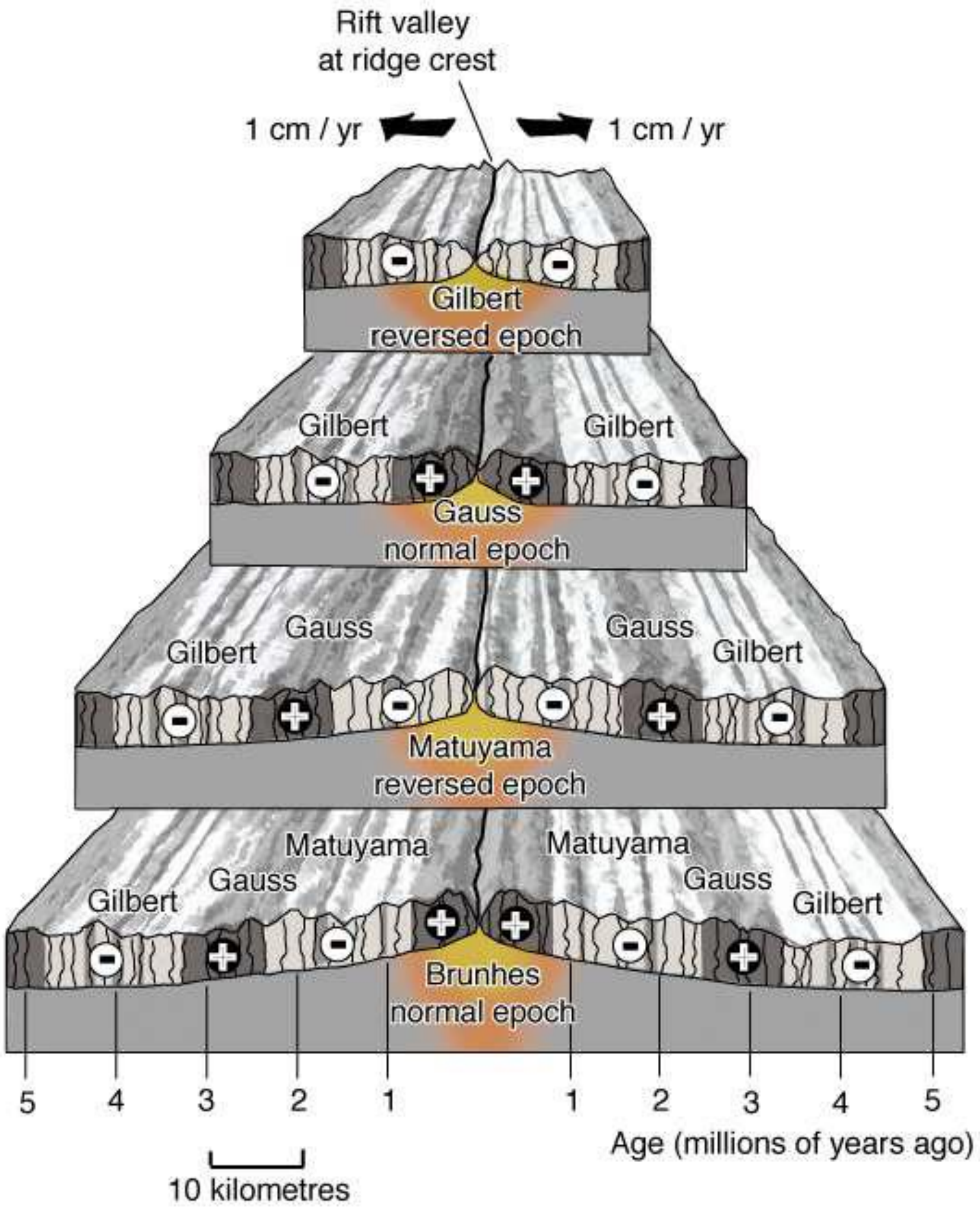




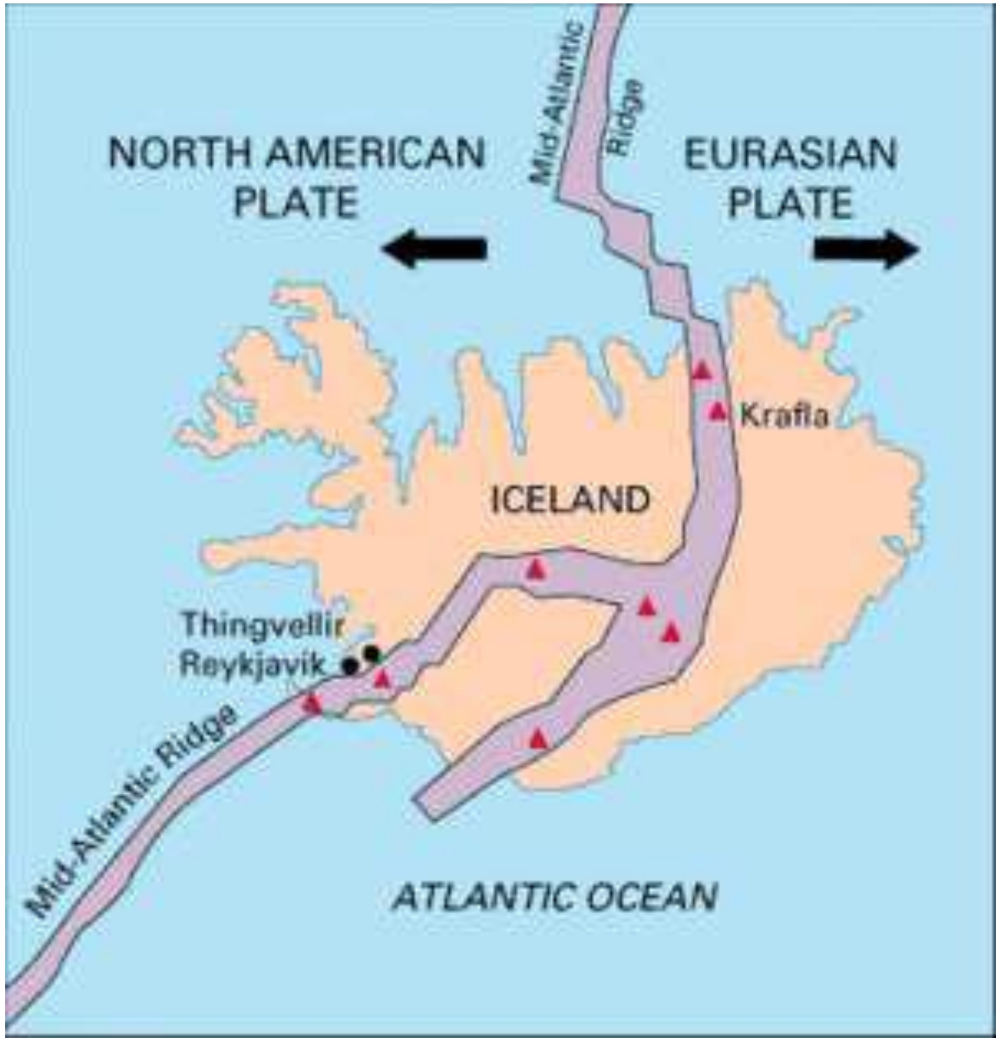
Fred Vine: 1965

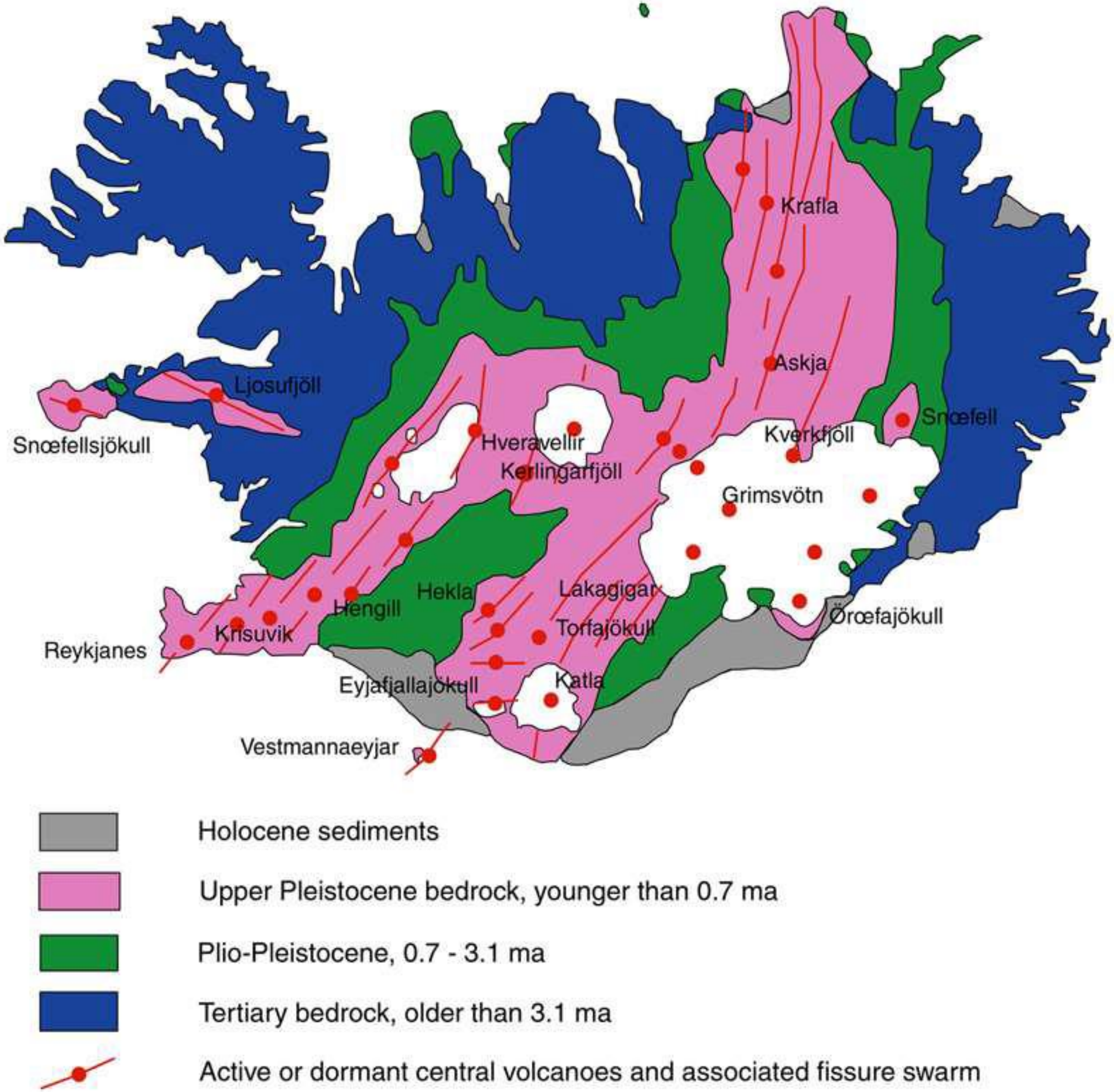
- **Magnetic stripes on the ocean floor**
- **At any given time, new seafloor records the magnetic polarity OF THAT TIME**
- **New seafloor pushes old seafloor off of the ridge = sea floor spreading**

"Vine-Morley-Mathews Hypothesis"



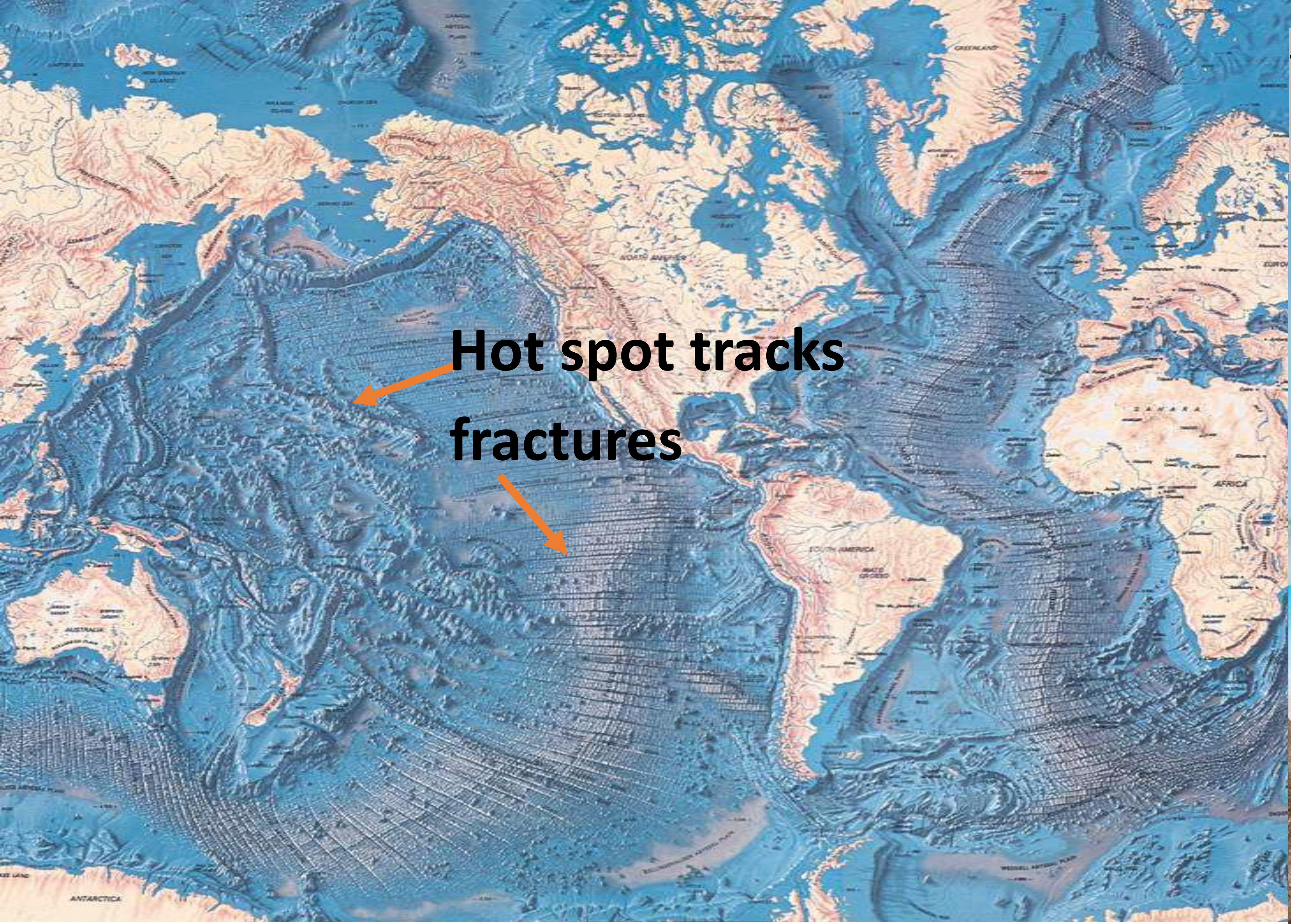
ICELAND:
How does the geology confirm sea floor spreading?

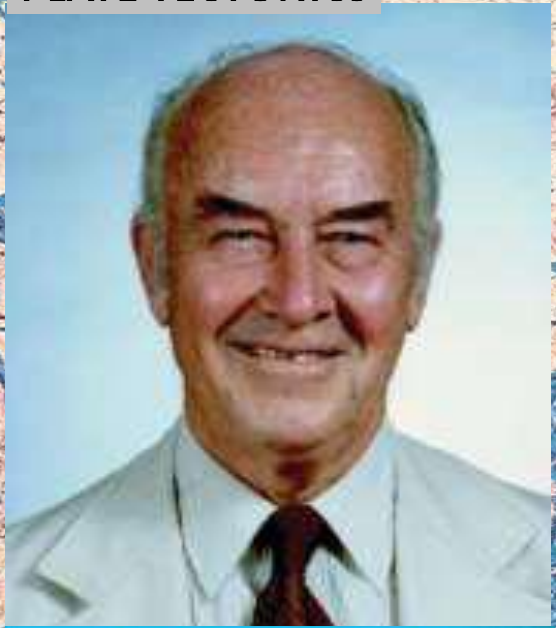


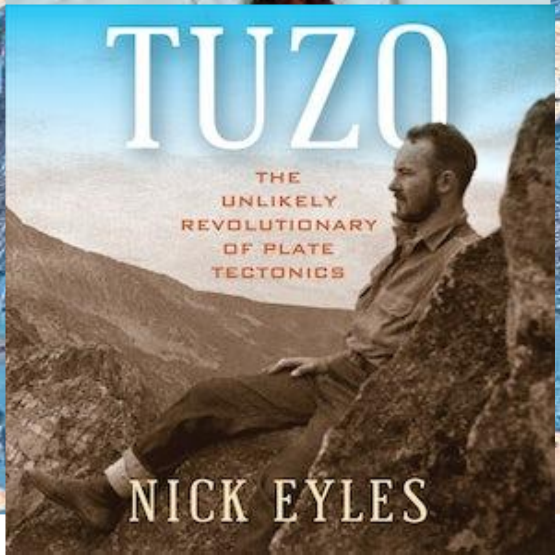



Direct measurement of plate movement by GPS
A fissure and a rift

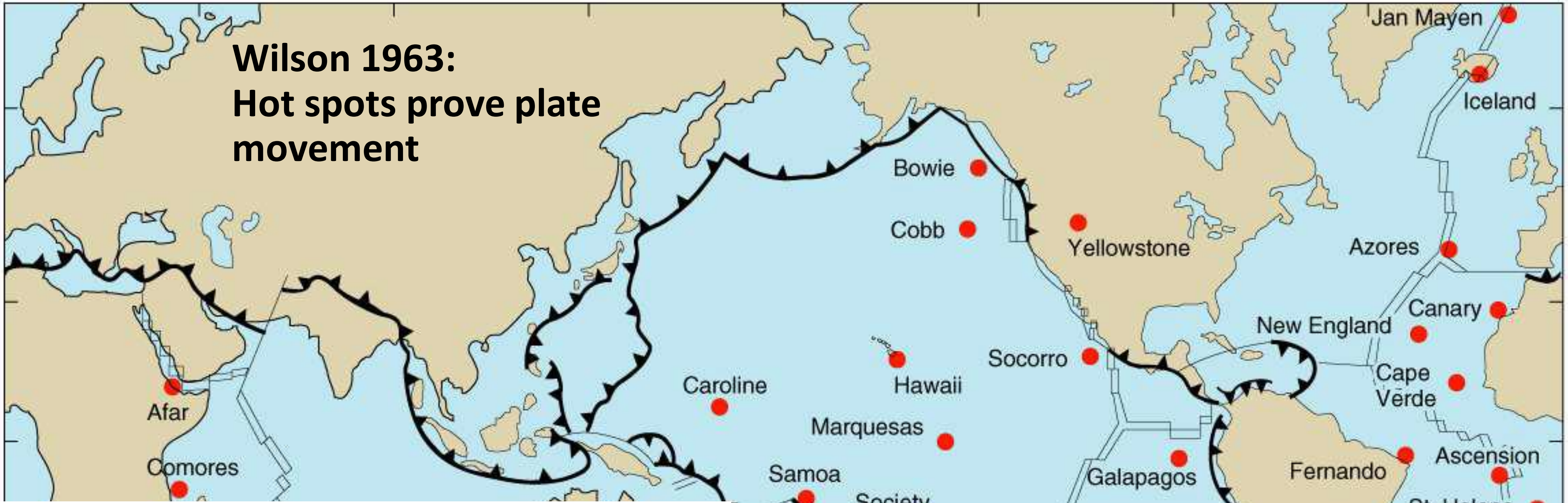
Jack Tuzo Wilson:
Professor of Geophysics at
U of T after 1946
‘PLATE TECTONICS’



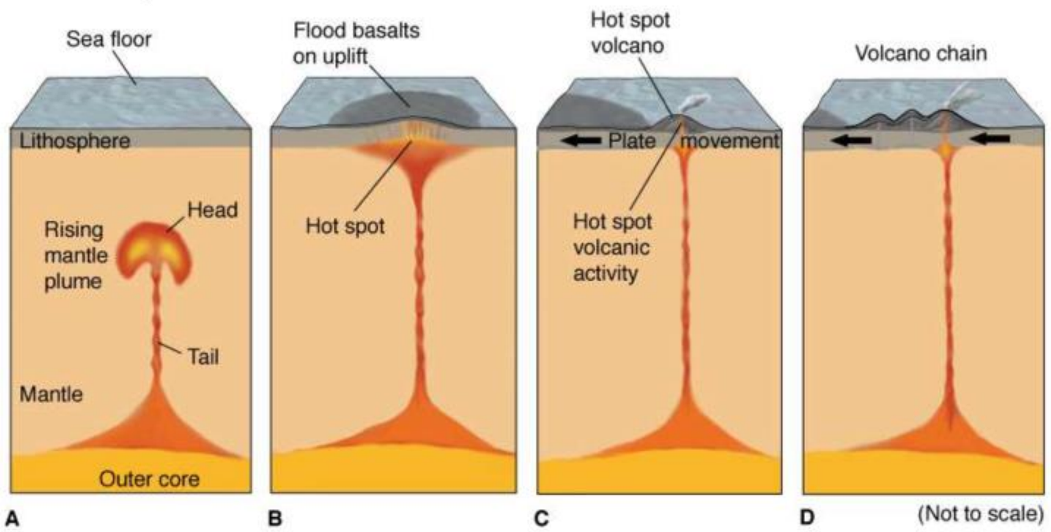


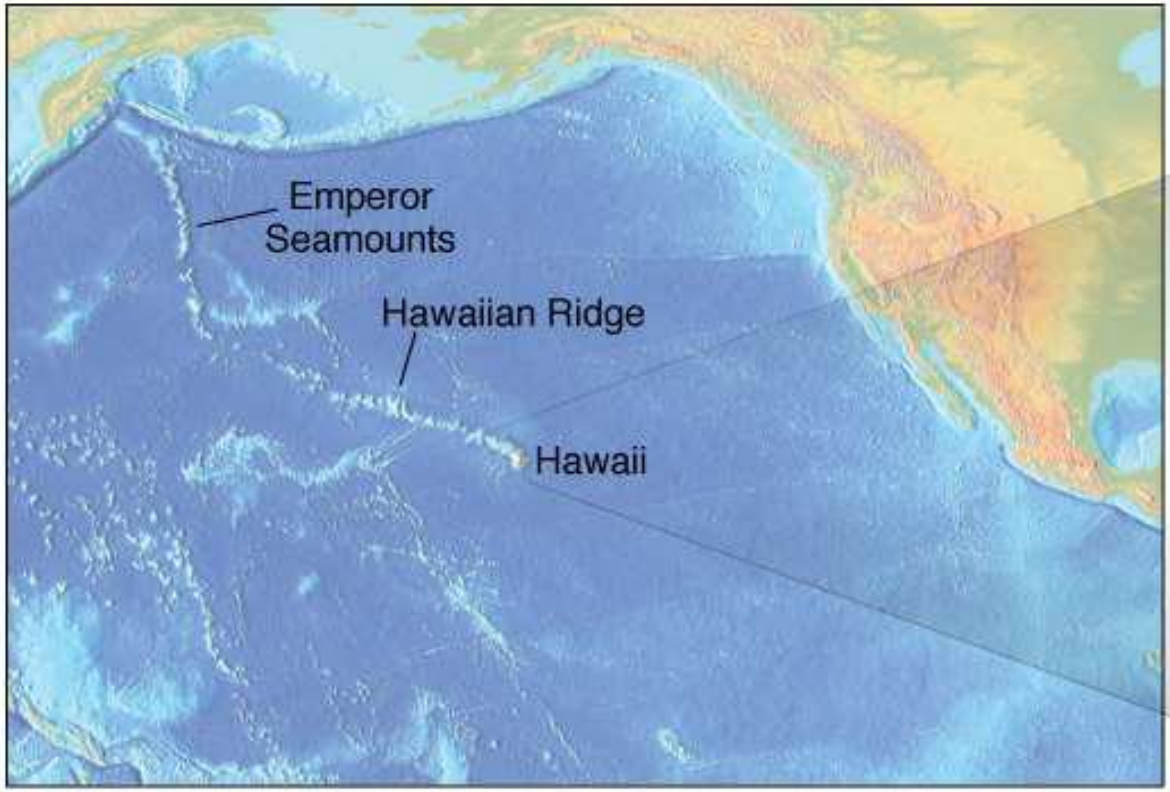


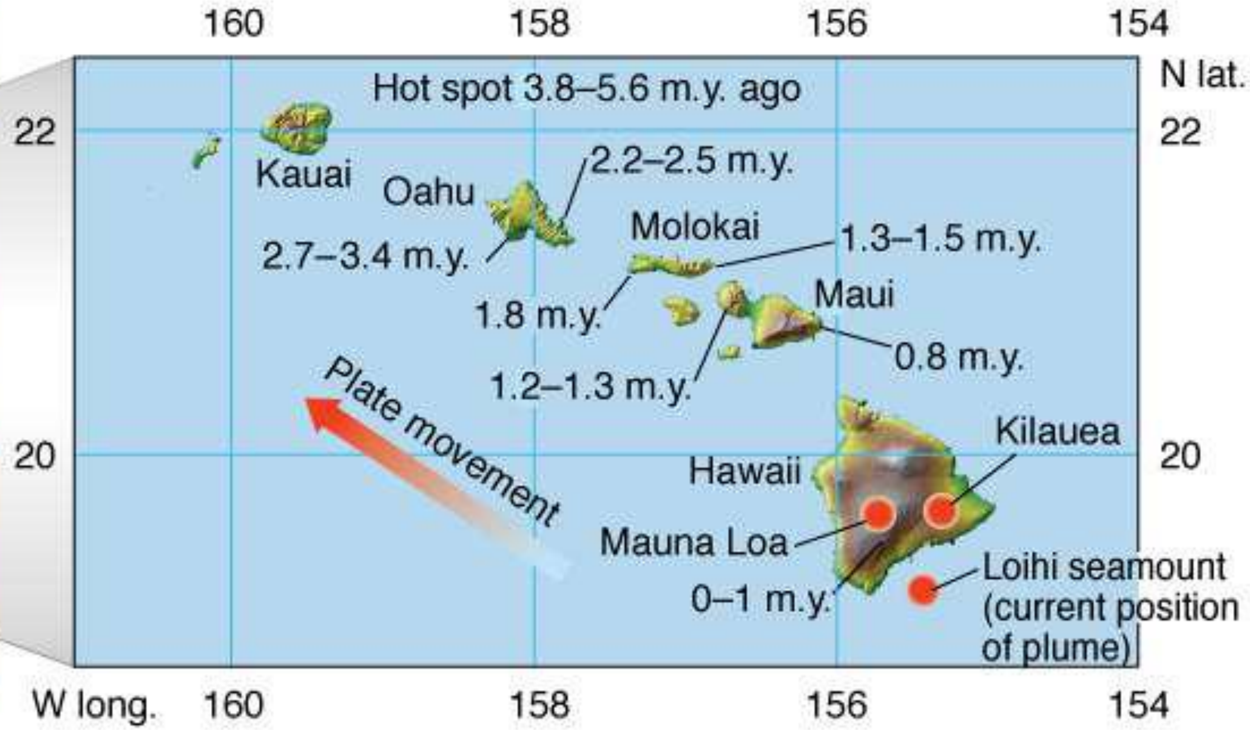
Wilson 1963: Hot spots prove plate movement

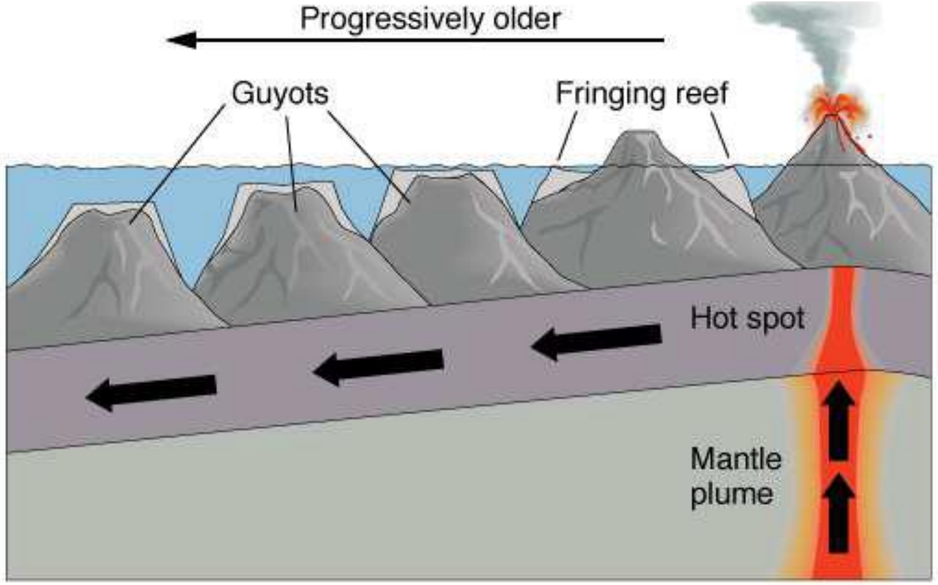


Hotspots



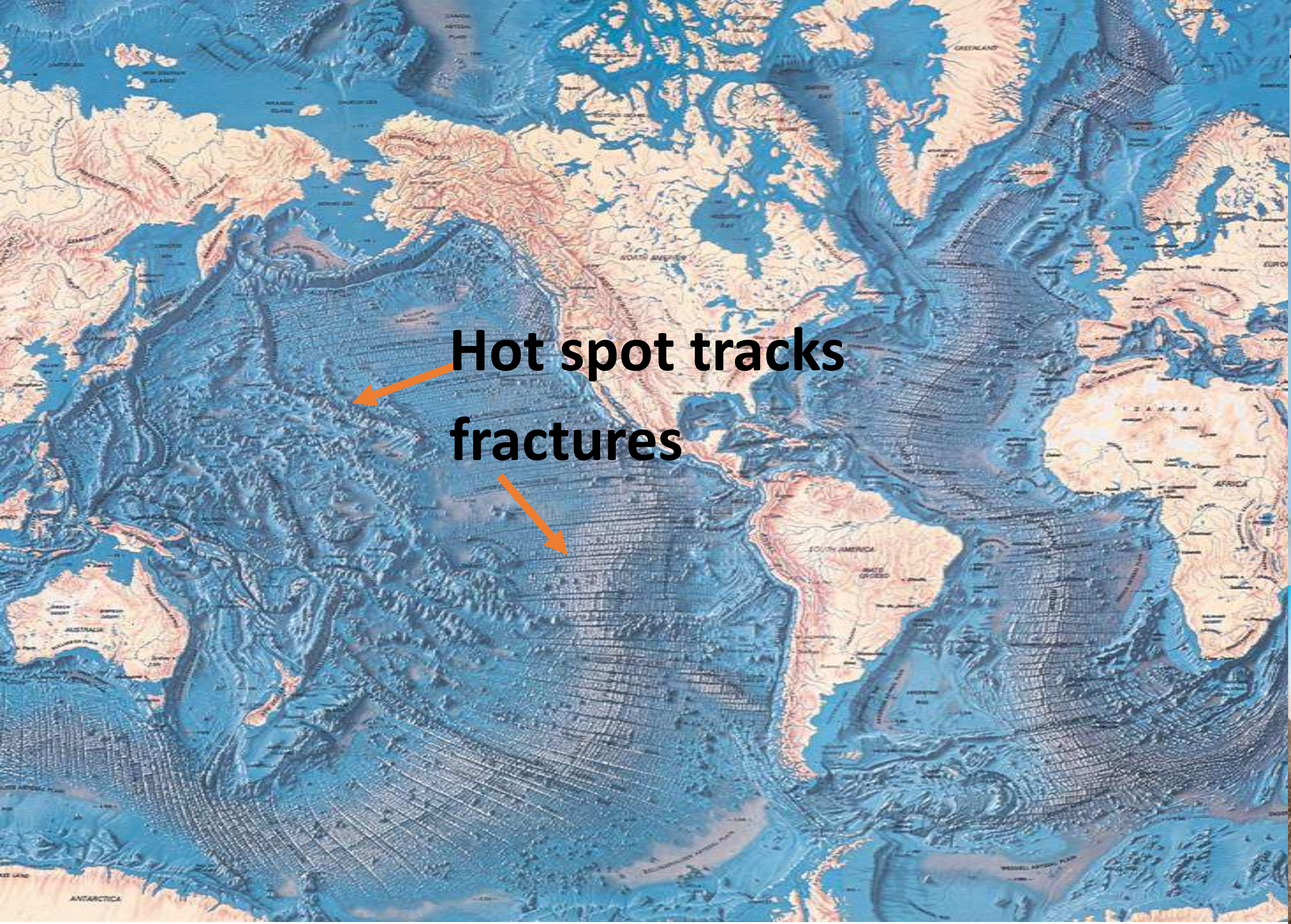

Emperor
Seamounts
Hawaiian Ridge
Hawaii

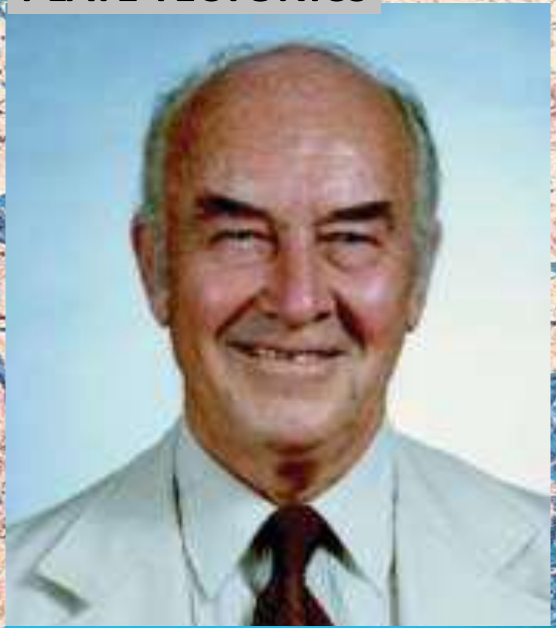

160
158
156
154
N lat.
22
20
Hot spot 3.8–5.6 m.y. ago
Kauai
Oahu
2.2–2.5 m.y.
2.7–3.4 m.y.
Molokai
1.3–1.5 m.y.
1.8 m.y.
Maui
0.8 m.y.
1.2–1.3 m.y.
Plate movement
Kilauea
Hawaii
Mauna Loa
Loihi seamount
(current position
of plume)
0–1 m.y.
W long.

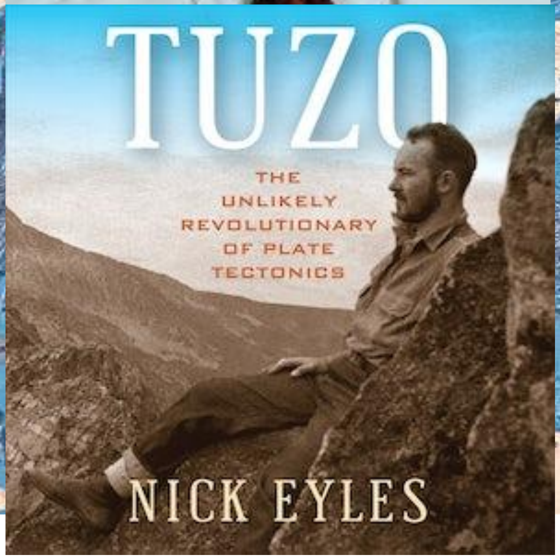

Progressively older
Guyots
Fringing reef
Hot spot
Mantle
plume

Atoll

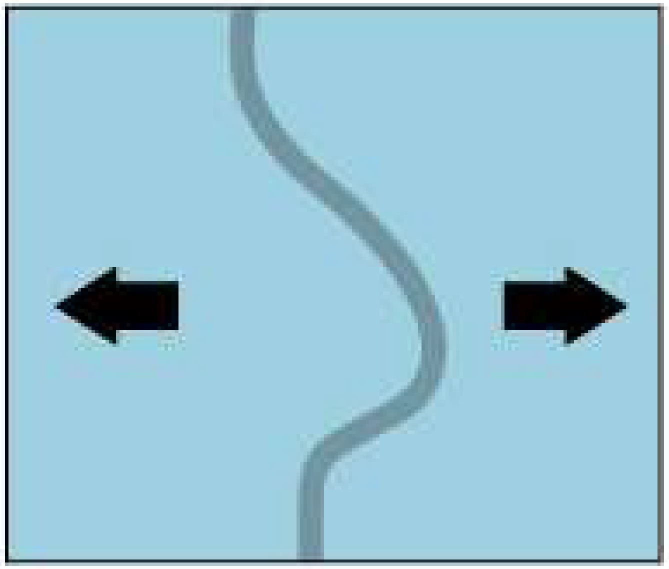
Jack Tuzo Wilson:
Professor of Geophysics at
U of T after 1946
'PLATE TECTONICS'



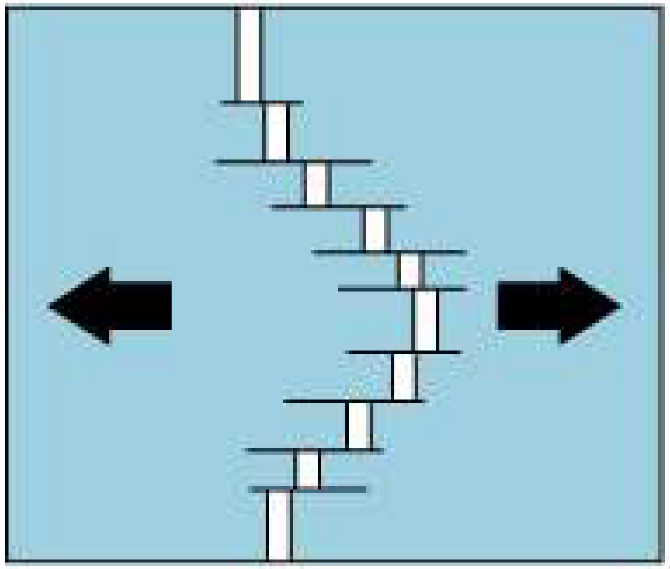


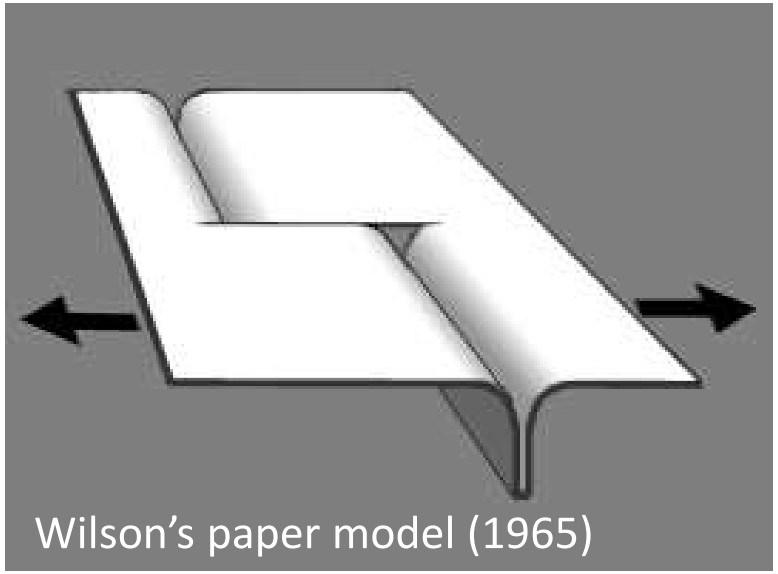


SINUOUS RIDGE

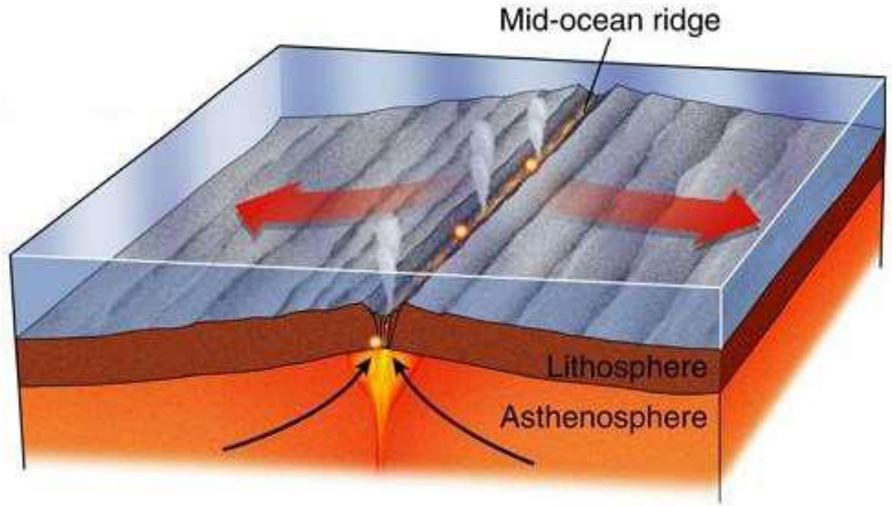


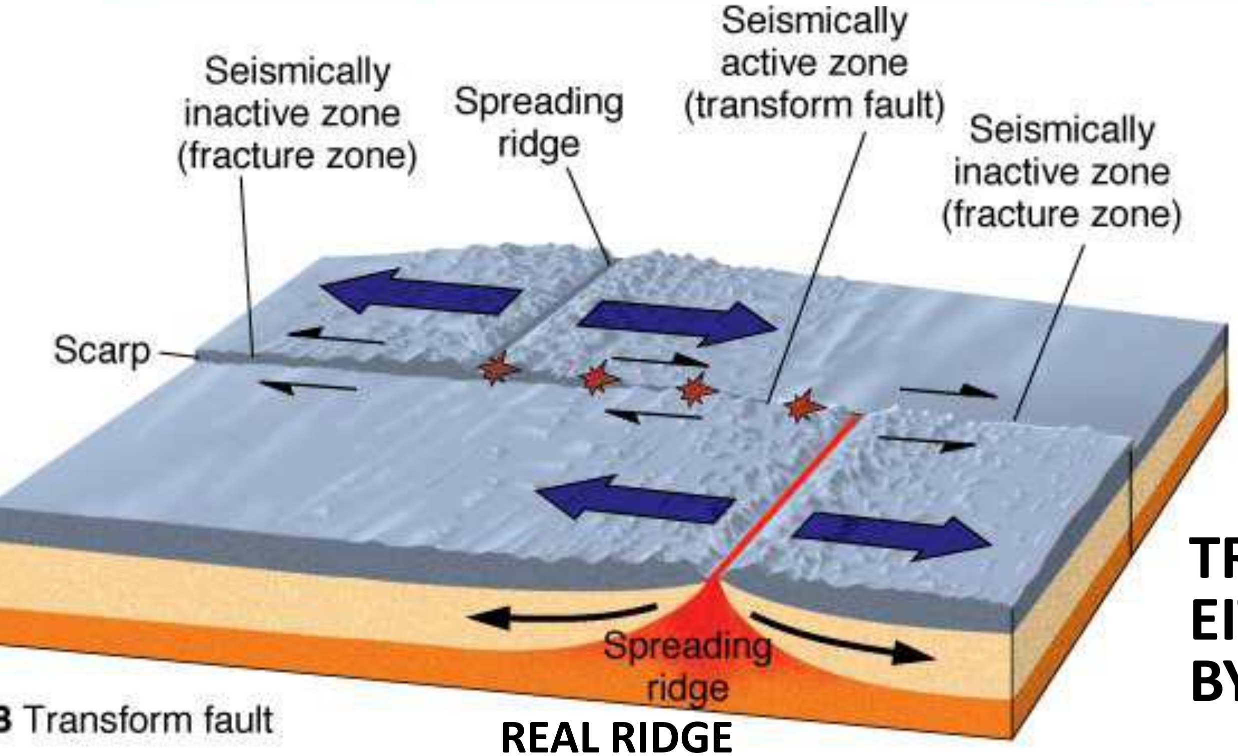
SEGMENTED RIDGE





SIMPLIFIED RIDGE



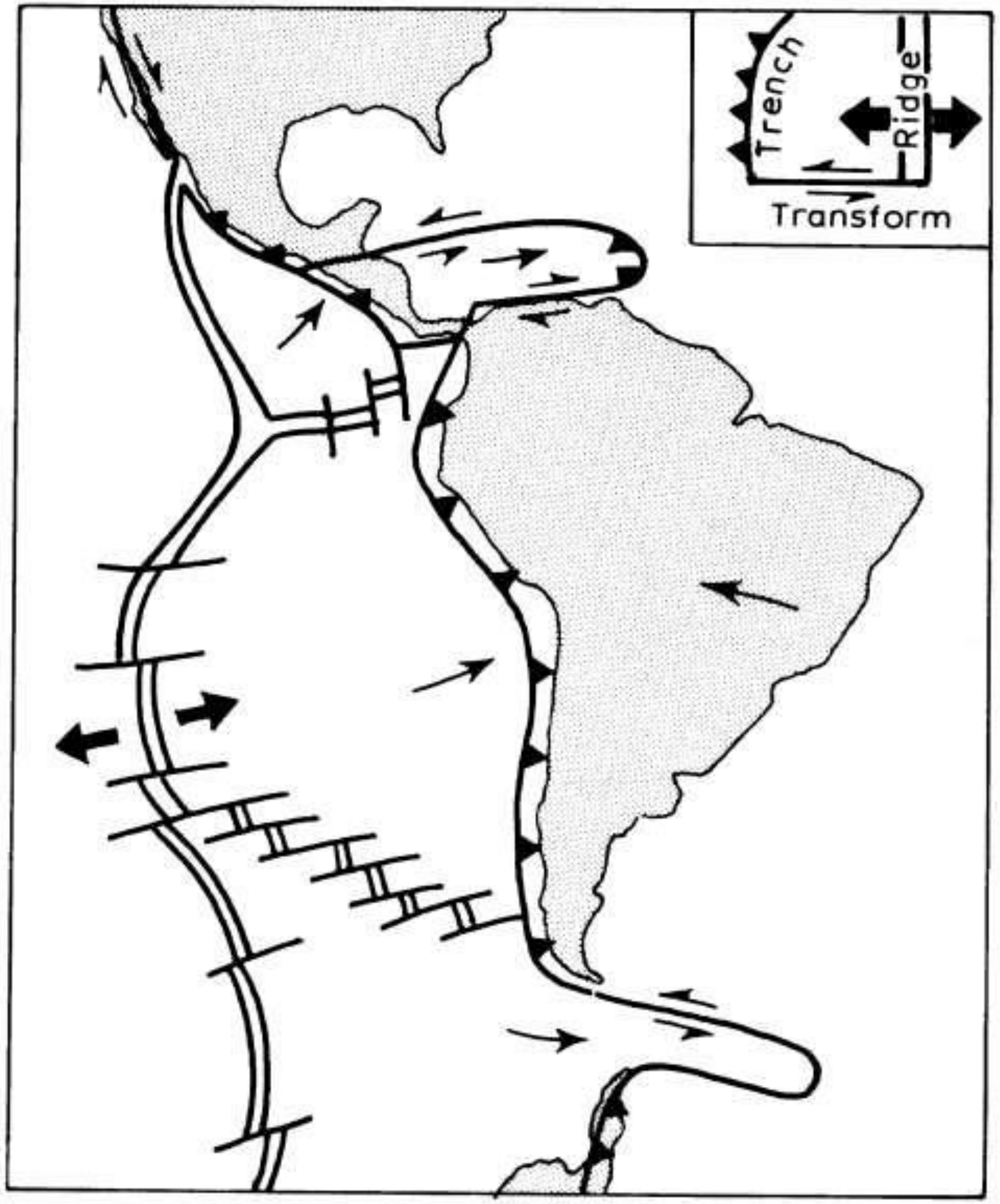


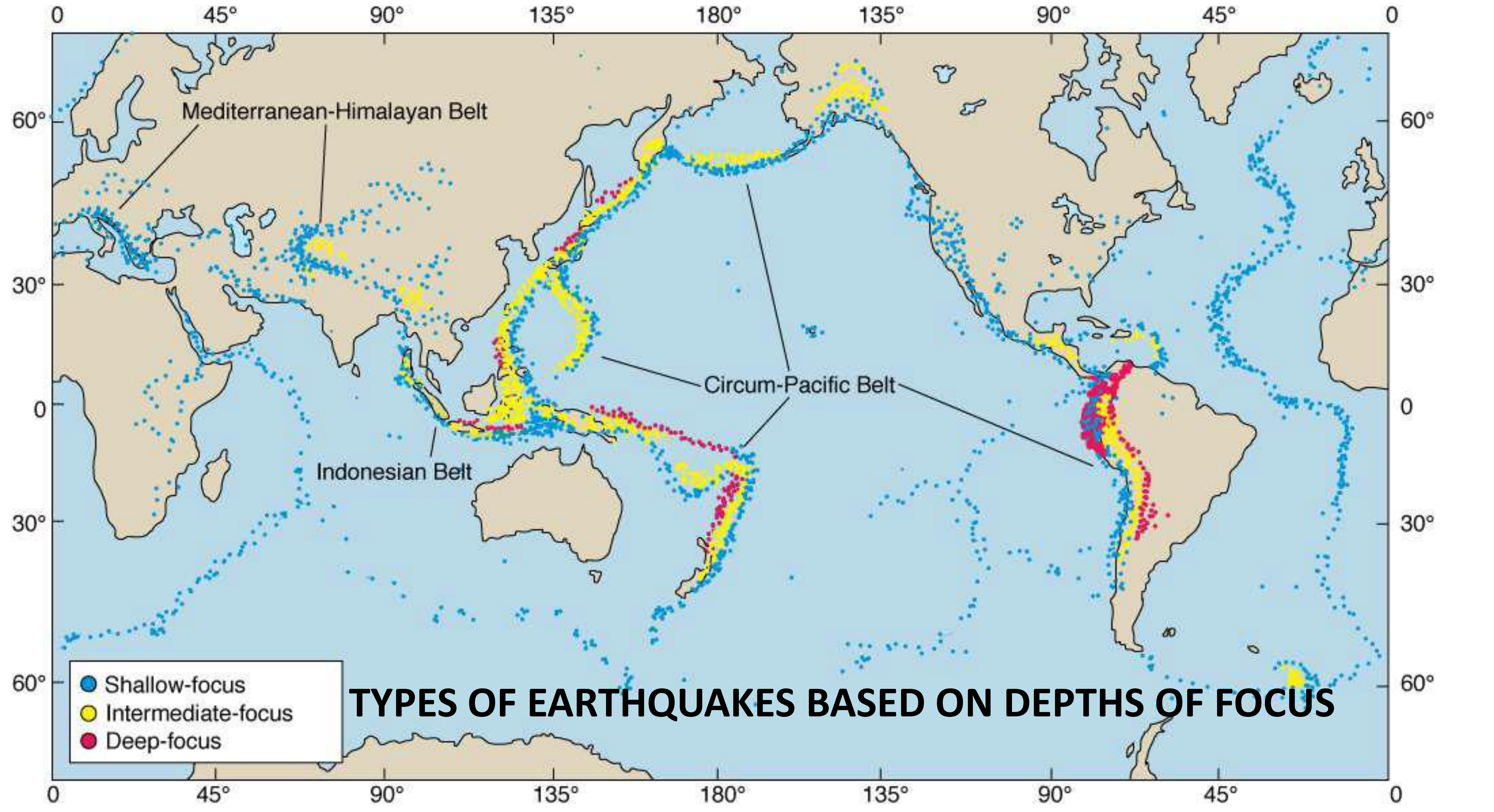
REAL RIDGE

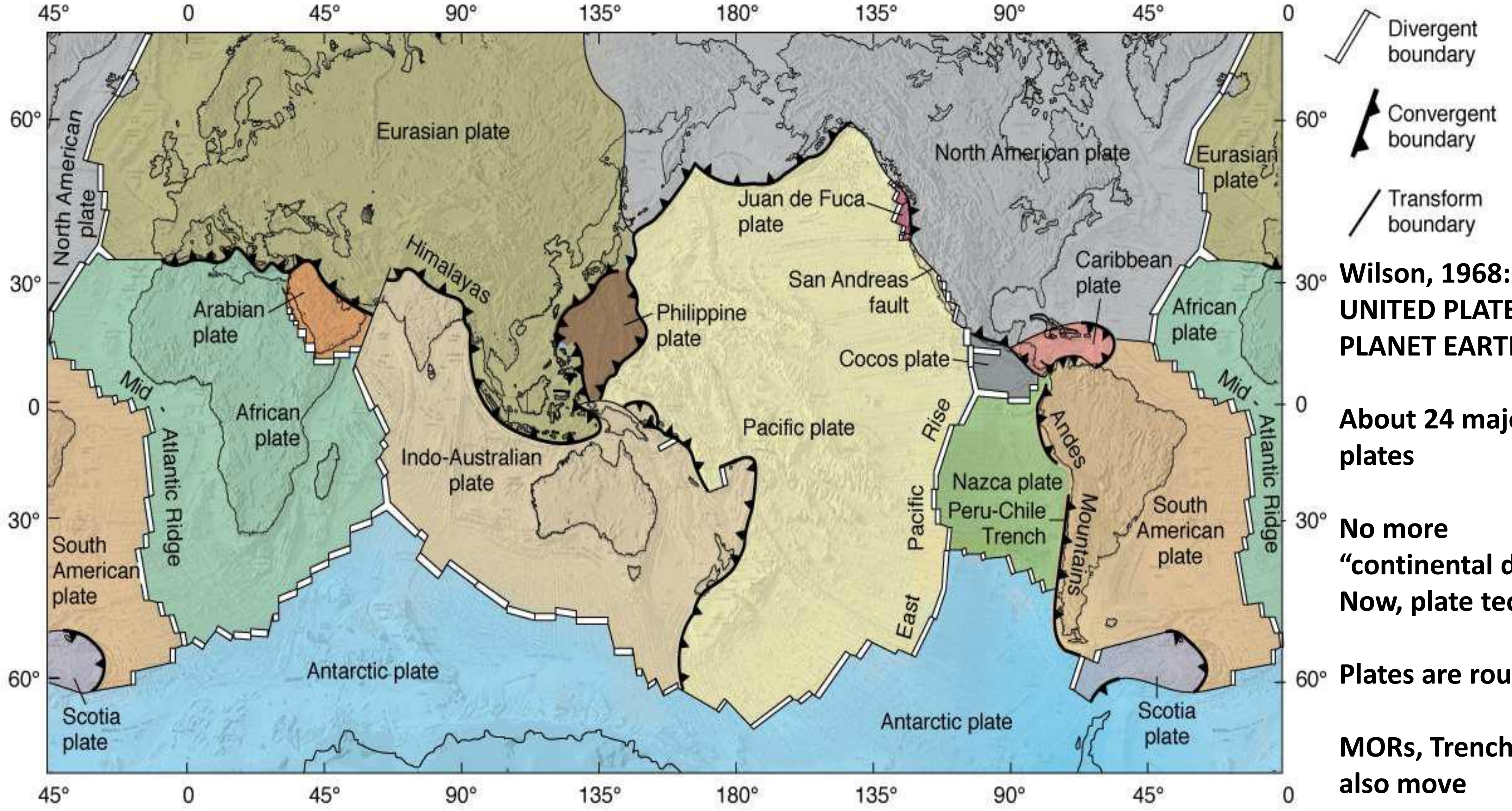
TRANSFORM FAULTS: ALLOW SPREADING EITHER SIDE OF SINUOUS MID-OCEAN RIDGES BY DIVIDING RIDGE INTO *SEGMENTS*

Wilson's second big contribution (1965): A third type of plate boundary

e.g. the San Andreas Fault




0
45°
90°
135°
180°
135°
90°
45°
0
60°
30°
0
30°
60°
Mediterranean-Himalayan Belt
Circum-Pacific Belt
Indonesian Belt
Shallow-focus
Intermediate-focus
Deep-focus
TYPES OF EARTHQUAKES BASED ON DEPTHS OF FOCUS



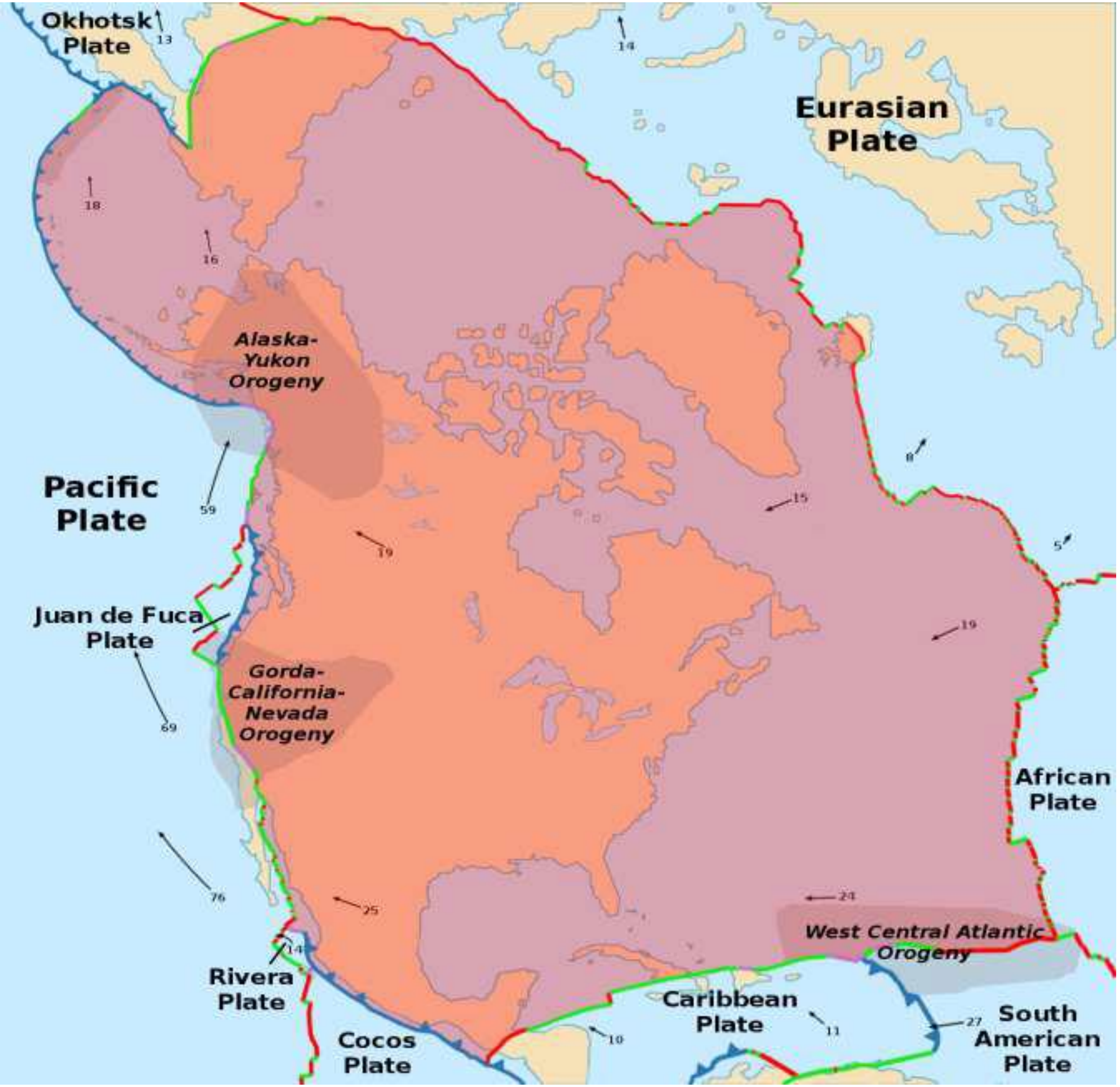
Wilson, 1968: THE UNITED PLATES OF PLANET EARTH

About 24 major plates

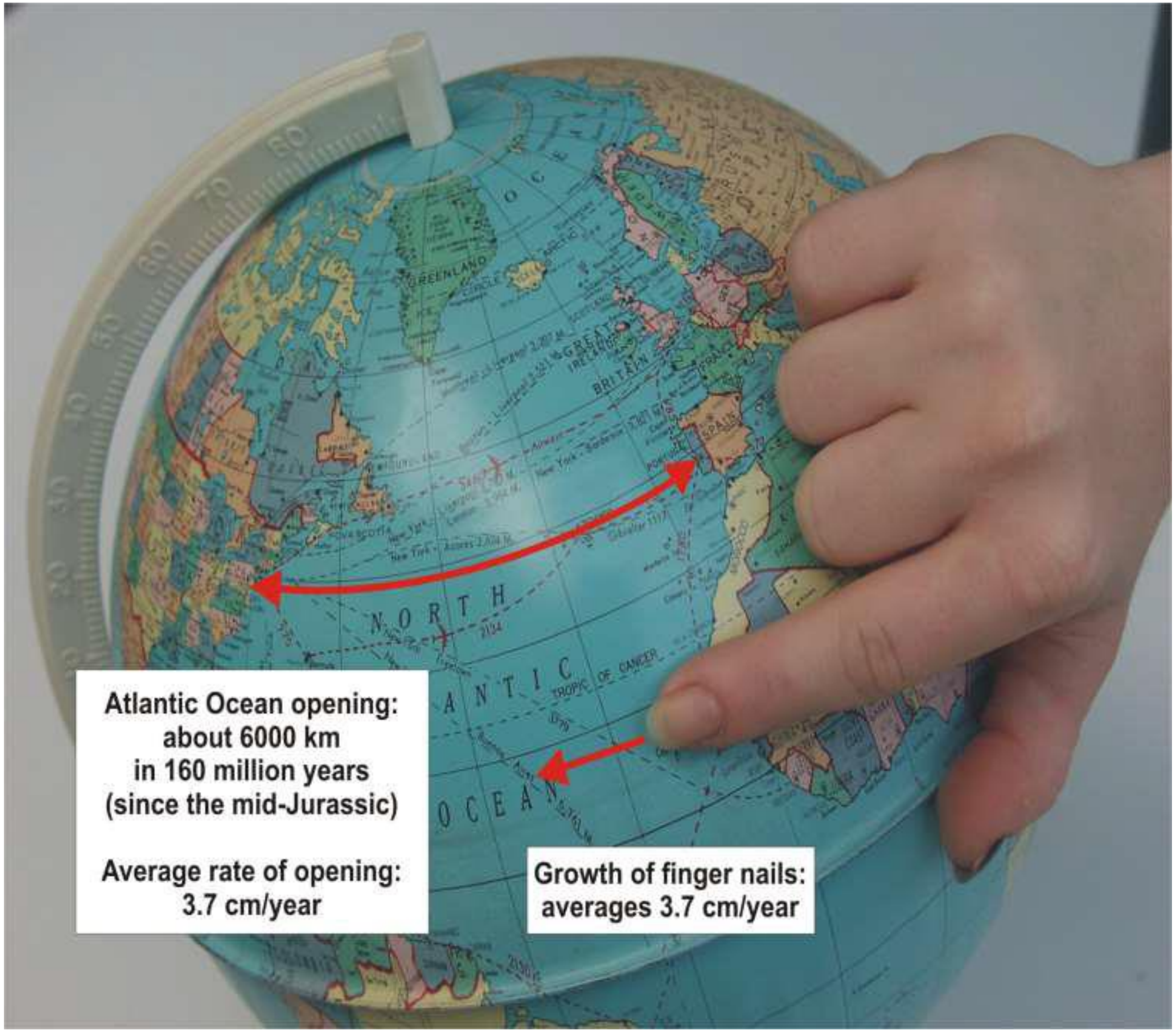
No more "continental drift"!! Now, plate tectonics

Plates are round

MORs, Trenches etc. also move



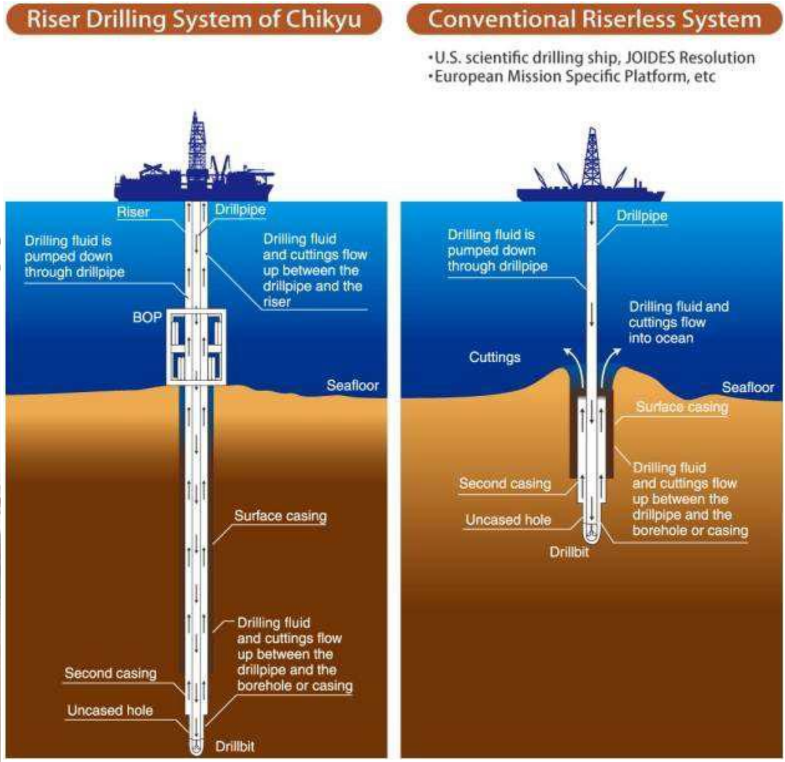
THE NORTH AMERICAN PLATE

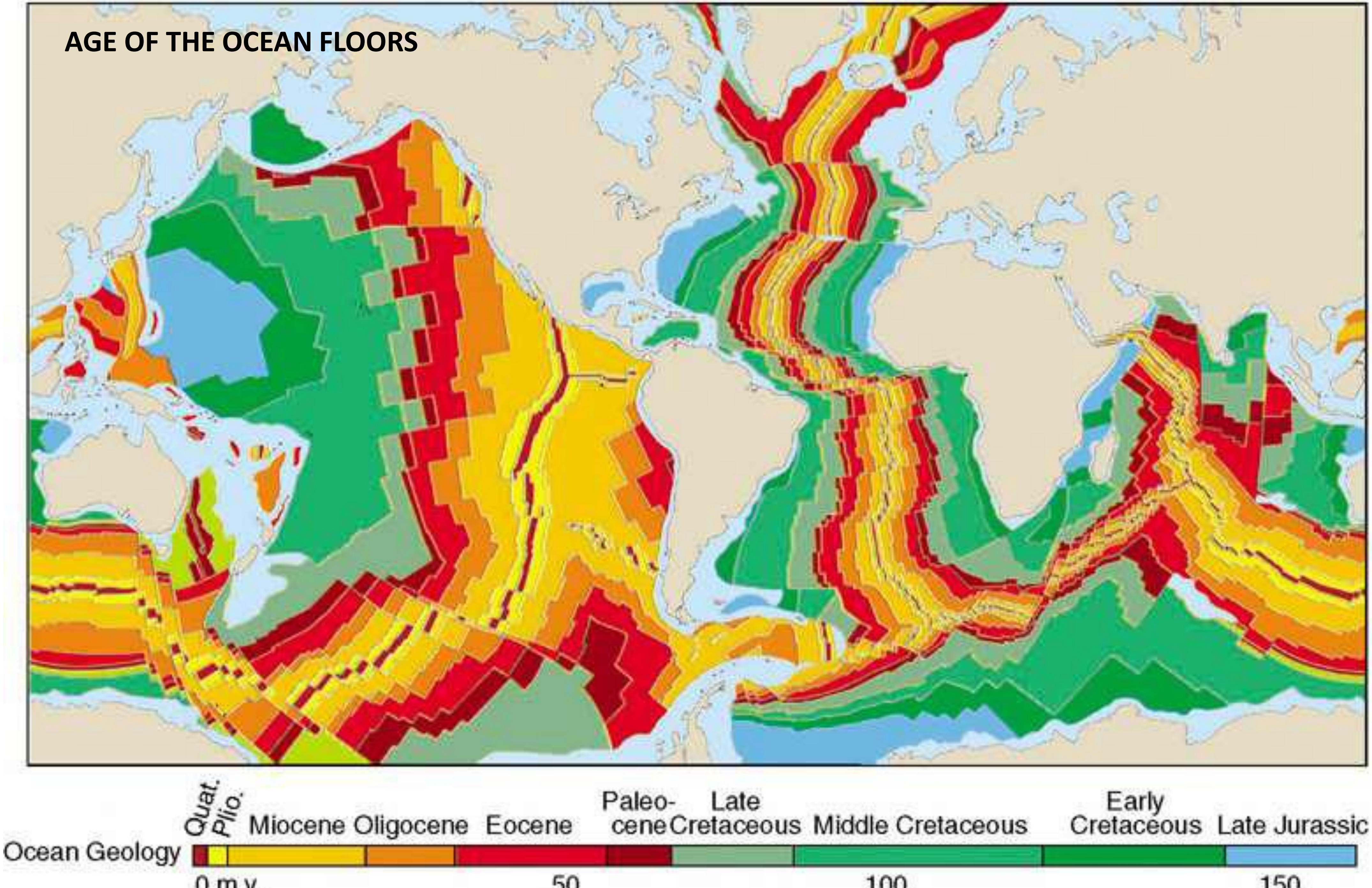

Atlantic Ocean opening:
about 6000 km
in 160 million years
(since the mid-Jurassic)
Average rate of opening:
3.7 cm/year
Growth of finger nails:
averages 3.7 cm/year

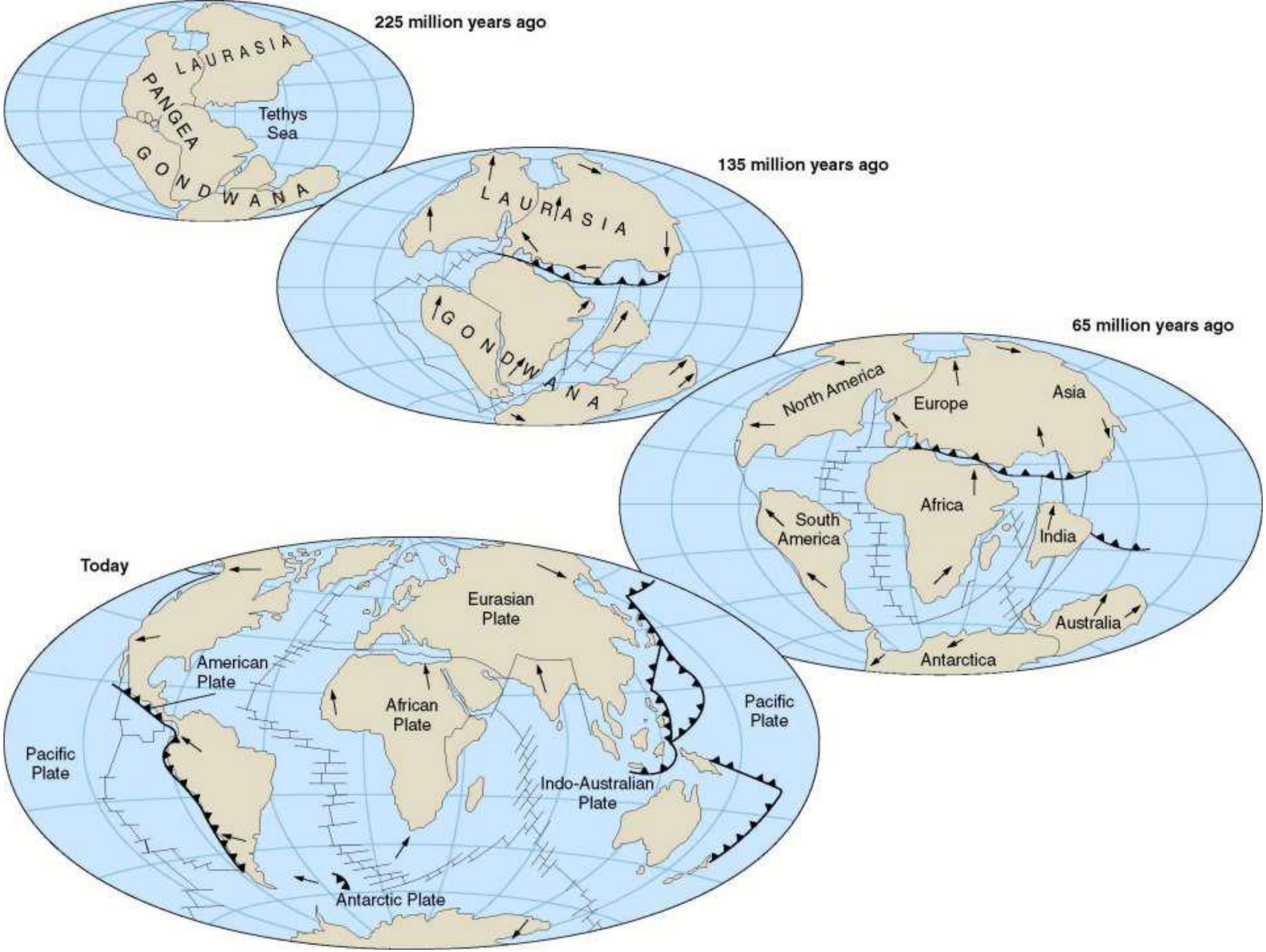
Groundtruthing Plate Tectonics by drilling the Ocean Floor

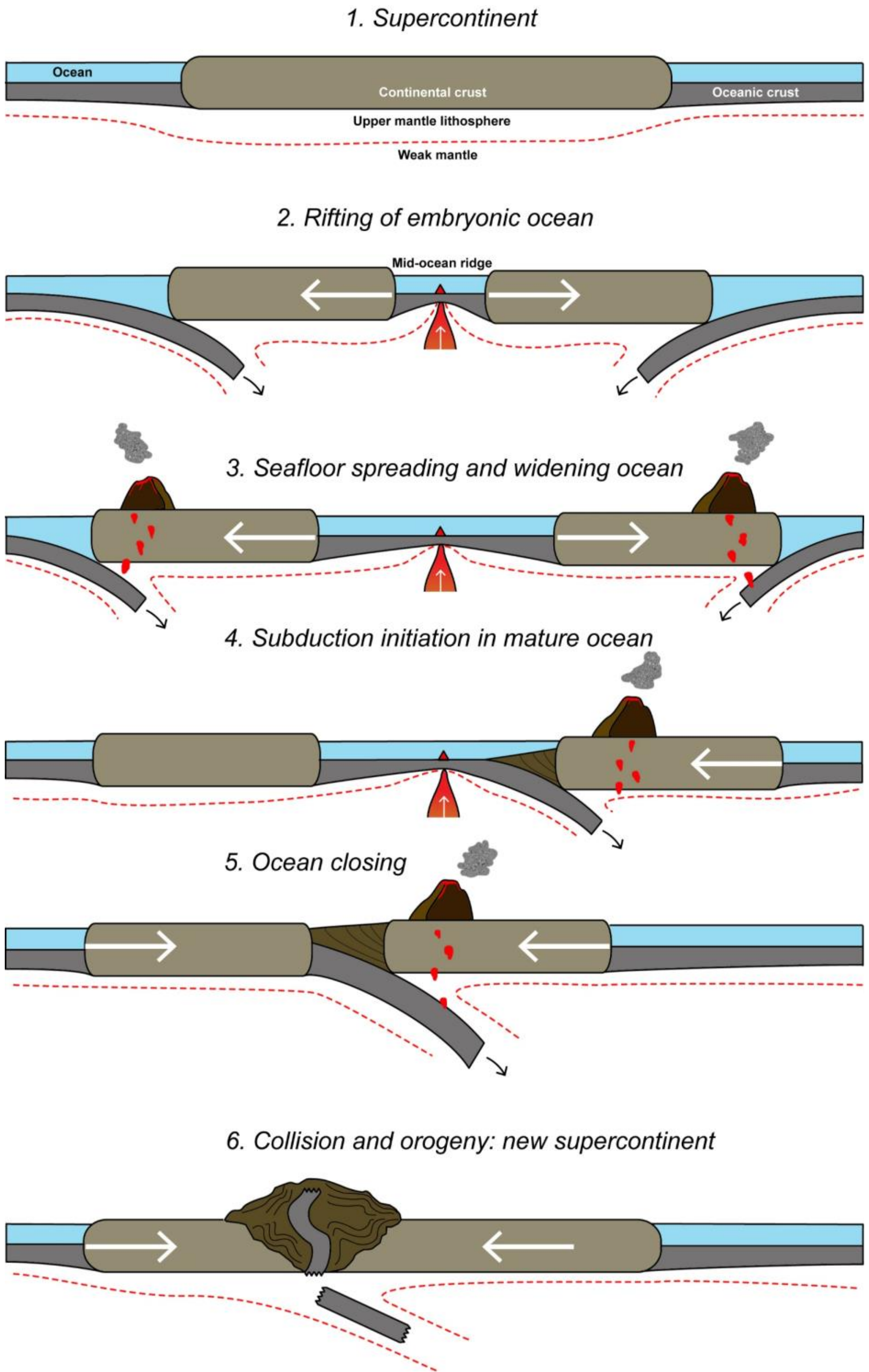
Resolution

Chikyu




AGE OF THE OCEAN FLOORS
Quat.
Plio.
Miocene
Oligocene
Eocene
Paleo-cene
Late Cretaceous
Middle Cretaceous
Early Cretaceous
Late Jurassic
Ocean Geology
0 m.y.
50
100
150


225 million years ago
LAURASIA
PANGEA
Tethys Sea
GONDWANA
135 million years ago
LAURASIA
GONDWANA
65 million years ago
North America
Europe
Asia
Africa
South America
India
Australia
Antarctica
Today
Eurasian Plate
American Plate
African Plate
Pacific Plate
Pacific Plate
Indo-Australian Plate
Antarctic Plate



Over the next few classes: THE WILSON CYCLE

- Approximately 500 million years in duration from beginning to end
- Last supercontinent was Pangea 250 million years ago (Ma: mega annum)
- “Oceans” are growing and dying all the time
  - Embryonic and juvenile oceans like the Red Sea
  - Mature oceans like the Atlantic
  - Dying Oceans like the Pacific
  - Dead Oceans like the Tethys